Elementary Analysis

Chapter 3: Sections 17–20

All 60 exercises, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 3. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Sections 17–20: Exercises 17.1–17.16, 18.1–18.12, 19.1–19.12, and 20.1–20.20.

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17 Continuous functions

Continuity is always relative to a function's domain. At a point x_0 in that domain, Definition 17.1 requires $f(x_n) \to f(x_0)$ for every domain sequence $x_n \to x_0$. The equivalent epsilon–delta condition of Theorem 17.2 tests only points x in the domain as well. This qualification matters for square roots, quotients, and isolated domain points.

Two simple starting functions are continuous on $\mathbb{R}$: the identity $I(x) = x$ and every constant function $C(x) = c$. Indeed, for $x_n \to x_0$, one has $I(x_n) = x_n \to x_0 = I(x_0)$ and $C(x_n) = c \to c = C(x_0)$. Theorems 17.3–17.5 then supply scalar multiples, sums, products, quotients with nonzero denominators, and compositions. Each application below checks the domain or sign condition it needs.

17.1 Exercise 17.1: Domains and values of two compositions

What we must show. For $f(x) = \sqrt{4 - x}$ on $(-\infty, 4]$ and $g(x) = x^2$ on $\mathbb{R}$, answer all four domain and composition questions.

Plan. A sum or product needs both original values. A composition also needs the inner value to belong to the outer function's domain.

(a) Find the domains before evaluating anything.

For $f + g$ and fg, intersect the two domains:

$$\operatorname{dom}(f + g) = \operatorname{dom}(fg) = (-\infty, 4] \cap \mathbb{R} = (-\infty, 4].$$

Their formulas there are $\sqrt{4 - x} + x^2$ and $x^2\sqrt{4 - x}$, respectively.

For $f \circ g$, the inner function g is defined for every real x, but its value must satisfy $g(x) \leq 4$. This means $x^2 \leq 4$, or equivalently $-2 \leq x \leq 2$. Hence

$$(f \circ g)(x) = \sqrt{4 - x^2}, \qquad \operatorname{dom}(f \circ g) = [-2, 2].$$

For $g \circ f$, first $x \leq 4$ is required to define $f(x)$. The value $f(x)$ is then real, and g accepts every real input. There is no additional restriction. Squaring the nonnegative square root gives

$$(g \circ f)(x) = (\sqrt{4 - x})^2 = 4 - x, \qquad \operatorname{dom}(g \circ f) = (-\infty, 4].$$

(b) Substitute the requested inputs into the appropriate formulas.

All three inputs $0, 1, 2$ lie in both composition domains. Their values, including the intermediate arithmetic, are

x	$(f \circ g)(x) = \sqrt{4 - x^2}$	$(g \circ f)(x) = 4 - x$
0	$\sqrt{4 - 0} = 2$	$4 - 0 = 4$
1	$\sqrt{4 - 1} = \sqrt{3}$	$4 - 1 = 3$
2	$\sqrt{4 - 4} = 0$	$4 - 2 = 2$

Thus the six values in the order requested are $2, 4, \sqrt{3}, 3, 0, 2$.

(c) Check equality as functions, not only equality of formulas.

The functions have different domains. Moreover, even on their common domain they differ: at 0 the first has value 2 and the second value 4. Either observation disproves equality.

(d) Test whether the inner value is admissible at 3.

Since $g(3) = 9 > 4$, the expression $f(g(3)) = f(9)$ is not defined as a real value. In contrast, $3 \leq 4$, so $f(3) = \sqrt{1} = 1$ is defined and $g(f(3)) = g(1) = 1$ is defined.

Conclusion. The sum and product domains are $(-\infty, 4]$; the two composition domains are $[-2, 2]$ and $(-\infty, 4]$. The compositions are unequal. At 3, only $g \circ f$ is meaningful.

17.2 Exercise 17.2: Compositions can remove or preserve a jump

What we must show. Let $f(x) = 0$ for $x < 0$, $f(x) = 4$ for $x \geq 0$, and $g(x) = x^2$ for every real x. Determine the four combinations in part (a), and decide which of the six functions in part (b) are continuous.

(a) Work separately on the two sides of zero.

Both original domains are $\mathbb{R}$. Sums and products are therefore defined on $\mathbb{R}$, with formulas

$$(f+g)(x) = \begin{cases} x^2, & x < 0, \\ x^2 + 4, & x \geq 0, \end{cases} \qquad (fg)(x) = \begin{cases} 0, & x < 0, \\ 4x^2, & x \geq 0. \end{cases}$$

For every real x, $g(x) = x^2 \geq 0$. Thus $f(g(x)) = 4$, including at $x = 0$, because $f(0) = 4$. Hence

$$(f \circ g)(x) = 4 \qquad (x \in \mathbb{R}).$$

For the opposite order, square the two possible values of f:

$$(g \circ f)(x) = \begin{cases} 0^2 = 0, & x < 0, \\ 4^2 = 16, & x \geq 0. \end{cases}$$

The composition domains are $\mathbb{R}$ as well, since all inner values belong to the outer domains.

(b) First check points other than zero.

At any $x_0 \neq 0$, a radius $|x_0|/2$ keeps x on the same side of zero as x_0. Each displayed function is then given near x_0 by one constant or polynomial formula. Such formulas are continuous by the identity, constant, sum, and product rules. Thus zero is the only possible discontinuity of any function in this exercise.

(b) Identify the three jumps with one sequence.

Take $x_n = -1/n$. Then $x_n \to 0$, but

$$f(x_n) = 0 \to 0 \neq 4 = f(0),$$

$$(f+g)(x_n) = 1/n^2 \to 0 \neq 4 = (f+g)(0),$$

and

$$(g \circ f)(x_n) = 0 \to 0 \neq 16 = (g \circ f)(0).$$

Definition 17.1 therefore makes all three functions discontinuous at zero.

(b) Check the functions that remain continuous at zero.

The function $g(x) = x^2$ is a product of two identity functions, so it is continuous everywhere. The function $f \circ g$ is constant, so it is continuous everywhere too.

For fg, its value at zero is 0, and for every real x, $|(fg)(x)| \leq 4x^2$. Given $\varepsilon > 0$, choose $\delta = \sqrt{\varepsilon}/2 > 0$. If $|x| < \delta$, then

$$|(fg)(x) - (fg)(0)| \leq 4x^2 < 4\delta^2 = \varepsilon.$$

Hence fg is continuous at zero and therefore on all of $\mathbb{R}$.

Conclusion. The globally continuous functions are g, fg, and $f \circ g$. The functions f, $f + g$, and $g \circ f$ are continuous away from zero but discontinuous at zero.

17.3 Exercise 17.3: Continuity from algebra and composition

What we must show. Prove continuity of all eight requested functions using the given familiar continuous functions and the section's rules.

Plan. Decompose each expression into known continuous maps, then check that every outer function receives an allowed input.

(a) The function $\log(1 + \cos^4 x)$.

The given continuity of cosine and repeated products show that $\cos^4 x$ is continuous on $\mathbb{R}$. Addition of the constant 1 preserves continuity. Furthermore, $1 + \cos^4 x \geq 1 > 0$ for every real x. The logarithm is continuous on $(0, \infty)$, which contains all these inner values. The composition theorem therefore proves continuity on the natural domain $\mathbb{R}$.

(b) The function $[\sin^2 x + \cos^6 x]^\pi$.

Products and a sum show that $u(x) = \sin^2 x + \cos^6 x$ is continuous on $\mathbb{R}$. Both summands are nonnegative. They cannot both vanish: if $\sin x = 0$, the identity $\sin^2 x + \cos^2 x = 1$ gives $\cos^2 x = 1$, so $\cos^6 x = 1$. If $\sin x \neq 0$, the first summand is positive instead. Thus $u(x) > 0$ for every x. The given power function $t \mapsto t^\pi$ is continuous for $t > 0$, so its composition with u is continuous on $\mathbb{R}$.

(c) The function 2^{x^2}.

The map $x \mapsto x^2 = I(x)I(x)$ is continuous on $\mathbb{R}$. The given map $t \mapsto 2^t$ is continuous for every real t. Its composition with x^2 is therefore continuous on $\mathbb{R}$.

(d) The function 8^x.

Use $8^x = 2^{3x}$. The inner map $x \mapsto 3x$ is continuous as a scalar multiple of the identity, and the outer map $t \mapsto 2^t$ is continuous on $\mathbb{R}$. Composition proves the claim on $\mathbb{R}$ without needing a new rule for the base 8.

(e) The tangent function on its stated domain.

Write $\tan x = \sin x / \cos x$. Sine and cosine are continuous on $\mathbb{R}$, and cosine is nonzero exactly away from $x = \pi/2 + k\pi$, $k \in \mathbb{Z}$. The quotient theorem proves continuity at every point of that natural domain.

(f) The function $x \sin(1/x)$ for $x \neq 0$.

The constant function 1 and identity are continuous; the quotient $r(x) = 1/x$ is therefore continuous on $D = \mathbb{R} \setminus \{0\}$. Since sine is continuous on all of $\mathbb{R}$, $\sin(r(x))$ is continuous on D by composition. Multiplying by the continuous identity gives continuity of $x \sin(1/x)$ on D.

(g) The function $x^2 \sin(1/x)$ for $x \neq 0$.

The sine factor is continuous on D by part (f)'s construction. The factor x^2 is continuous on $\mathbb{R}$, hence on D. Their product is continuous at every point of D.

(h) The function $x^{-1} \sin(1/x^2)$ for $x \neq 0$.

On D, $x^2 \neq 0$. The quotient $1/x^2$ is continuous there, and its composition with sine is continuous there. The other factor $1/x$ is continuous on D too. Their product is the requested continuous function.

Conclusion. Every function is continuous on its natural stated domain: $\mathbb{R}$ in (a)–(d), the nonzero-cosine domain in (e), and $\mathbb{R} \setminus \{0\}$ in (f)–(h). No value at zero is asserted in the last three parts.

17.4 Exercise 17.4: Continuity of the square root including its endpoint

What we must show. Prove that $f(x) = \sqrt{x}$ is continuous on $[0, \infty)$.

Plan. An estimate valid even at zero avoids dividing by a possibly small sum of square roots.

Prove the estimate for any two domain points.

If $x \geq y \geq 0$, then $\sqrt{x} - \sqrt{y} \geq 0$ and $\sqrt{x} + \sqrt{y} \geq \sqrt{x} - \sqrt{y}$. Multiplication by $\sqrt{x} - \sqrt{y}$ preserves this order, giving

$$x - y = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq (\sqrt{x} - \sqrt{y})^2.$$

Taking nonnegative square roots yields $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. If $y > x$, interchange the two variables; the same absolute-value inequality results. Thus it holds for every $x, y \geq 0$.

Choose delta at an arbitrary point of the domain.

Fix $x_0 \geq 0$ and $\varepsilon > 0$, and put $\delta = \varepsilon^2 > 0$. For a domain point $x \geq 0$ with $|x - x_0| < \delta$, the estimate gives

$$|\sqrt{x} - \sqrt{x_0}| \leq \sqrt{|x - x_0|} < \sqrt{\delta} = \varepsilon.$$

This is Theorem 17.2. It includes $x_0 = 0$, because the condition tests only $x \geq 0$ and never requires a negative input.

Conclusion. The square-root function is continuous at every point of its domain, including zero.

17.5 Exercise 17.5: Powers and polynomials are continuous

What we must show. Prove part (a) for every positive integer power, then deduce part (b) for every polynomial.

(a) Build powers by induction using the product rule.

For $m = 1$, $x \mapsto x^m = x$ is the identity, already shown continuous. Suppose $x \mapsto x^m$ is continuous on $\mathbb{R}$. Then $x^{m+1} = x^m x$ is the product of that continuous function and the identity. Theorem 17.4(ii) makes it continuous on $\mathbb{R}$. Mathematical induction proves continuity for all $m \in \mathbb{N}$.

(b) Assemble a polynomial from finitely many continuous terms.

Let

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n, \qquad a_0, \ldots, a_n \in \mathbb{R}.$$

The term a_0 is constant and continuous. For $j \geq 1$, part (a) makes x^j continuous, and Theorem 17.3 makes $a_j x^j$ continuous, including when $a_j = 0$. Theorem 17.4(i) preserves continuity under addition of two terms. Applying it finitely many times proves continuity of the displayed sum at every real point. A constant polynomial is already covered even if there are no positive-power terms.

Conclusion. Every positive integer power and every real polynomial is continuous on $\mathbb{R}$.

17.6 Exercise 17.6: Continuity of rational functions on their domains

What we must show. Let p, q be polynomial functions and define

$$D = \{x \in \mathbb{R} : q(x) \neq 0\}, \qquad f(x) = p(x)/q(x) \quad (x \in D).$$

Prove f is continuous.

Check the hypotheses at an arbitrary allowed point.

Fix $x_0 \in D$. Exercise 17.5 makes both p and q continuous at x_0, and membership in D gives $q(x_0) \neq 0$. The quotient theorem, Theorem 17.4(iii), therefore applies and gives continuity of p/q at x_0.

For a direct reading of that theorem, take any $x_n \in D$ with $x_n \to x_0$. Continuity gives $p(x_n) \to p(x_0)$ and $q(x_n) \to q(x_0) \neq 0$. The quotient limit law gives

$$f(x_n) = \frac{p(x_n)}{q(x_n)} \longrightarrow \frac{p(x_0)}{q(x_0)} = f(x_0).$$

Every denominator in this display is nonzero by the domain conditions. Since $x_0 \in D$ was arbitrary, the function is continuous on D.

Conclusion. A rational function is continuous wherever its denominator does not vanish. Excluded zeros of the denominator are not points at which continuity of this domain function is tested.

17.7 Exercise 17.7: Linear maps, absolute value, and scalar multiples

What we must show. Prove (a) continuity of $x \mapsto kx$, (b) continuity of absolute value, and (c) Theorem 17.3 by composition.

(a) A linear map is a polynomial.

For a fixed real k, the function $G(t) = kt$ is a polynomial with constant coefficient zero and linear coefficient k. Exercise 17.5 makes it continuous on $\mathbb{R}$. This includes $k = 0$, in which case it is constant.

(b) Bound the error in absolute value by the input error.

The triangle inequality gives $|x| \leq |x - x_0| + |x_0|$, so $|x| - |x_0| \leq |x - x_0|$. Interchanging x and x_0 gives $|x_0| - |x| \leq |x - x_0|$. Together these two inequalities are

$$\big||x| - |x_0|\big| \leq |x - x_0|\,.$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $|x - x_0| < \delta$ forces $||x| - |x_0|| < \varepsilon$. Thus $H(t) = |t|$ is continuous at every real point.

(c) Compose with the given function.

Suppose f is continuous at a point x_0 of its domain. The outer maps G and H are continuous at the real number $f(x_0)$ by parts (a) and (b). Theorem 17.5 gives continuity at x_0 of

$$(G \circ f)(x) = kf(x), \qquad (H \circ f)(x) = |f(x)|\,.$$

The outer maps have domain $\mathbb{R}$, so both compositions retain the whole domain of f. These are precisely the two functions in Theorem 17.3.

Conclusion. Linear maps and absolute value are continuous on $\mathbb{R}$; composition with them gives a second proof of Theorem 17.3.

17.8 Exercise 17.8: Formulas and continuity for a pointwise minimum

What we must show. Verify both pointwise identities and use one to prove the continuity assertion in part (c).

(a) Verify the absolute-value formula by cases.

For arbitrary real numbers a, b, suppose first that $a \le b$. Then $|a - b| = b - a$, and

$$\frac{a + b - |a - b|}{2} = \frac{a + b - (b - a)}{2} = a = \min\{a, b\}.$$

If $a > b$, then $|a - b| = a - b$, and the same expression is $(a + b - (a - b))/2 = b = \min\{a, b\}$. Substituting $a = f(x)$ and $b = g(x)$ wherever both are defined gives

$$\min(f, g) = \tfrac{1}{2}(f + g) - \tfrac{1}{2}\,|f - g|\,.$$

(b) Verify the negated-maximum formula.

If $a \le b$, negation reverses order, so $-a \ge -b$. Thus $\max\{-a, -b\} = -a$ and its negative is a. If $a > b$, one has $-a < -b$, so the negative of the maximum is b. In both cases, $\min\{a, b\} = -\max\{-a, -b\}$. Pointwise substitution therefore gives $\min(f, g) = -\max(-f, -g)$.

(c) Apply the continuity rules to part (a).

Suppose f and g are continuous at x_0 and both are defined there. Their sum and difference are continuous at x_0 by the sum and scalar rules. The absolute-value rule makes $|f - g|$ continuous there. Multiplying the two terms of part (a) by $1/2$ and $-1/2$, then adding, preserves continuity. The resulting function is $\min(f, g)$. Its domain is the common domain of f and g, so the point x_0 is an allowed point throughout the argument.

Conclusion. Both identities hold, and the pointwise minimum of two functions continuous at x_0 is continuous there.

17.9 Exercise 17.9: Four explicit epsilon–delta proofs

What we must show. Verify the epsilon–delta condition directly in each part, with a stated positive choice of δ.

(a) The square function at $x_0 = 2$.

Fix $\varepsilon > 0$. Factor the error:

$$\left|x^2 - 4\right| = |x - 2|\,|x + 2|\,.$$

To bound the second factor, first require $|x - 2| < 1$. Then $|x + 2| \le |x - 2| + 4 < 5$. Choose

$$\delta = \min\{1, \varepsilon/5\} > 0.$$

If $|x - 2| < \delta$, both preliminary restrictions hold, and $|x^2 - 4| \le 5\,|x - 2| < 5\delta \le \varepsilon$. This proves continuity at 2.

(b) The square-root function at zero.

The domain restriction is $x \ge 0$. Given $\varepsilon > 0$, put $\delta = \varepsilon^2 > 0$. If $x \ge 0$ and $|x - 0| < \delta$, then $0 \le x < \varepsilon^2$ and

$$\left|\sqrt{x} - \sqrt{0}\right| = \sqrt{x} < \varepsilon.$$

The proof tests only domain points, as required at this endpoint.

(c) The function $x\sin(1/x)$ extended by zero.

Let $f(0) = 0$ and $f(x) = x\sin(1/x)$ for $x \ne 0$. For nonzero x, the bound $|\sin(1/x)| \le 1$ gives $|f(x)| \le |x|$. It also holds at $x = 0$, where both sides are zero. Given $\varepsilon > 0$, choose $\delta = \varepsilon$. If $|x| < \delta$, then

$$|f(x) - f(0)| = |f(x)| \le |x| < \varepsilon.$$

Hence f is continuous at zero despite the oscillating sine factor.

(d) The cube function at an arbitrary $x_0 \in \mathbb{R}$.

Fix $\varepsilon > 0$ and set $M = |x_0| + 1 > 0$. If $|x - x_0| < 1$, then $|x| \le |x - x_0| + |x_0| < M$ and $|x_0| < M$. Thus

$$\left|x^2 + xx_0 + x_0^2\right| \le |x|^2 + |x|\,|x_0| + |x_0|^2 \le 3M^2.$$

Using the hinted factorization, choose

$$\delta = \min\left\{1, \frac{\varepsilon}{3M^2}\right\} > 0.$$

For $|x - x_0| < \delta$, the local bound applies, and

$$\left|x^3 - x_0^3\right| = |x - x_0|\left|x^2 + xx_0 + x_0^2\right| \le 3M^2\,|x - x_0| < 3M^2\delta \le \varepsilon.$$

Conclusion. The stated delta choices verify continuity in all four parts directly, with all local bounds imposed before use.

17.10 Exercise 17.10: Three explicit sequences witnessing discontinuity

What we must show. Show discontinuity at zero in parts (a)–(c) by finding domain sequences approaching zero whose function values do not approach the assigned value zero.

(a) The one-sided step function.

Here $f(x) = 1$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$. Take $x_n = 1/n$. Then $x_n > 0$ and $x_n \to 0$, but $f(x_n) = 1$ for all n. The image sequence tends to 1, whereas $f(0) = 0$. This violates Definition 17.1.

(b) The oscillating sine function.

Here $g(x) = \sin(1/x)$ for $x \neq 0$ and $g(0) = 0$. Put

$$x_n = \frac{1}{2\pi n + \pi/2}.$$

The denominator is positive and tends to $+\infty$, so $x_n > 0$ and $x_n \to 0$. At these inputs,

$$g(x_n) = \sin(2\pi n + \pi/2) = 1.$$

Thus $g(x_n)$ tends to 1, not to $g(0) = 0$.

(c) The signum function.

For $\operatorname{sgn}(x) = 1$ at positive inputs, -1 at negative inputs, and 0 at zero, again take $x_n = 1/n$. Then $x_n \to 0$ while $\operatorname{sgn}(x_n) = 1$ for all n, contrary to the image limit required by continuity.

Conclusion. Each function is discontinuous at zero. In every case the displayed sequence stays in the domain and approaches the point, while its image remains at distance 1 from the value there.

17.11 Exercise 17.11: Testing continuity using only monotone sequences

What we must show. Prove that f is continuous at a domain point x_0 if and only if every monotone domain sequence $x_n \to x_0$ satisfies $f(x_n) \to f(x_0)$.

Plan. For the reverse implication, a failure of continuity produces a bad sequence; Theorem 11.4 supplies a monotone subsequence of that sequence without repairing its bad images.

Forward implication: restrict the sequences already tested.

If f is continuous at x_0, Definition 17.1 gives the image limit for every domain sequence approaching x_0. In particular it gives it for every monotone such sequence.

Reverse implication: express failure of continuity precisely.

Assume the stated condition for monotone sequences, and suppose for contradiction that f is not continuous at x_0. The negation of Theorem 17.2 supplies one fixed $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there is u in the domain with

$$|u - x_0| < \delta, \qquad |f(u) - f(x_0)| \geq \varepsilon_0.$$

For each $n \in \mathbb{N}$, use $\delta = 1/n$ and select such a point u_n. Then $|u_n - x_0| < 1/n$ proves $u_n \to x_0$, but every image has error at least ε_0.

Select monotone inputs while retaining their bad images.

Theorem 11.4 gives a monotone subsequence (u_{n_k}). It still lies in the domain, and Theorem 11.3 gives $u_{n_k} \to x_0$. The assumed monotone-sequence criterion would therefore give $f(u_{n_k}) \to f(x_0)$. That is impossible, since

$$|f(u_{n_k}) - f(x_0)| \geq \varepsilon_0 \qquad \text{for every } k.$$

For example, no image can satisfy the eventual error bound $\varepsilon_0/2$. The contradiction proves continuity.

Conclusion. Monotone test sequences suffice. The argument selects a monotone subsequence; it does not assume the original sequence of bad inputs is monotone.

17.12 Exercise 17.12: A continuous function is determined on a dense set

What we must show. Prove (a) that zero values at all rational points of (a, b) force zero values everywhere, and (b) that equality of two continuous functions at those rational points forces equality throughout the interval.

(a) Approximate an arbitrary interval point by rational points inside it.

Fix $x \in (a, b)$. For each $n \in \mathbb{N}$, put

$$\rho_n = \min\{(x - a)/2, (b - x)/2, 1/n\} > 0.$$

The open interval $(x - \rho_n, x + \rho_n)$ lies in (a, b) and has positive length. Rational density supplies $r_n \in \mathbb{Q}$ in that interval. In particular, $|r_n - x| < \rho_n \leq 1/n$, so $r_n \to x$. Since f is continuous at x and $f(r_n) = 0$ for every n,

$$f(x) = \lim_{n\to\infty} f(r_n) = \lim_{n\to\infty} 0 = 0.$$

The point x was arbitrary, so f vanishes everywhere on (a, b).

(b) Reduce equality to the zero-valued case.

The difference $h = f - g$ is continuous on (a, b) by the sum and scalar rules. At every rational point r in that interval, the hypothesis gives $h(r) = f(r) - g(r) = 0$. Part (a), applied to h, yields $h(x) = 0$ for every $x \in (a, b)$. Rearranging $f(x) - g(x) = 0$ gives $f(x) = g(x)$ there.

Conclusion. Both assertions follow by taking limits along rational points that stay inside the common domain.

17.13 Exercise 17.13: Rational and irrational inputs with different values

What we must show. Prove (a) that the rational-indicator function is discontinuous everywhere, and (b) that the function equal to x at rational inputs and zero at irrational inputs is continuous exactly at zero.

Preparation: obtain both kinds of inputs near any real point.

Rationals are dense in $\mathbb{R}$. Irrationals are dense as well: given $u < v$, rational density gives a rational r in $(u - \sqrt{2}, v - \sqrt{2})$. Then $r + \sqrt{2} \in (u, v)$ is irrational, since rationality of that sum would make $\sqrt{2}$ the difference of two rationals.

For each real a and each n, choose a rational r_n and an irrational t_n in $(a - 1/n, a + 1/n)$. Both sequences tend to a, since their distances from a are less than $1/n$.

(a) The indicator takes incompatible limits along these inputs.

Let $f(x) = 1$ for rational x and $f(x) = 0$ for irrational x. At the two sequences just chosen, $f(r_n) = 1$ and $f(t_n) = 0$ for every n. If f were continuous at a, both image sequences would have to tend to $f(a)$. Their limits are 1 and 0, which are different. Thus f cannot be continuous at a. As a was arbitrary, it is discontinuous everywhere.

(b) At zero, the magnitude of the function is small with the input.

Let $h(x) = x$ for rational x and $h(x) = 0$ for irrational x. Then $h(0) = 0$, and in both cases $|h(x)| \le |x|$. Given $\varepsilon > 0$, choose $\delta = \varepsilon$. For $|x| < \delta$,

$$|h(x) - h(0)| \le |x| < \varepsilon.$$

Hence h is continuous at zero.

(b) At a nonzero point, the two dense input sets still disagree.

Fix $a \neq 0$ and choose the rational and irrational sequences from the preparation step. Then

$$h(r_n) = r_n \longrightarrow a, \qquad h(t_n) = 0 \longrightarrow 0.$$

Continuity at a would require both limits to equal $h(a)$. Because $a \neq 0$, the two limits differ, giving a contradiction.

Conclusion. The function in (a) is nowhere continuous. The function in (b) is continuous at zero and at no other point.

17.14 Exercise 17.14: A denominator-sensitive function on the rationals

What we must show. For nonzero rational $x = p/q$ in lowest terms with $q > 0$, let $f(x) = 1/q$; also let $f(0) = 1$ and let f vanish on irrationals. Prove continuity at every irrational point and discontinuity at every rational point.

Plan. Near an irrational point, only finitely many fractions in a bounded neighborhood have small denominators. Avoid those finitely many fractions; all remaining nonzero values of f are small.

Unify the notation at the rational point zero.

Write $0 = 0/1$ in lowest terms. The same denominator rule then gives $1/q = 1 = f(0)$. Thus all rational inputs, including zero, can be discussed with a positive reduced denominator.

Continuity at an irrational point: isolate the finitely many exceptions.

Fix $a \in \mathbb{R} \setminus \mathbb{Q}$ and $\varepsilon > 0$. Choose an integer $Q > \max\{1, 1/\varepsilon\}$, so $1/Q < \varepsilon$. Consider

$$F = \left\{ \frac{p}{q} \in [a - 1, a + 1] : p \in \mathbb{Z},\ 1 \le q \le Q,\ p/q \text{ is in lowest terms} \right\}.$$

This set is finite. Indeed, for a fraction in this set,

$$|p| = q\,|p/q| \le q(|a| + 1) \le Q(|a| + 1).$$

There are only finitely many possible integers p in this fixed bounded range, and only Q possible denominators. Their possible pairs, and hence the resulting fractions, are finite. The argument concerns a bounded neighborhood; fractions with small denominators are not a finite set on the whole real line.

No member of F equals a, because a is irrational. If $F \neq \varnothing$, let $d = \min\{|r - a| : r \in F\} > 0$ and set $\delta = \min\{1, d/2\}$. If $F = \varnothing$, set $\delta = 1$ instead. In either case $\delta > 0$.

Check every nearby input, including rational ones.

Suppose $|x - a| < \delta$. Then $x \in [a - 1, a + 1]$, and $x \notin F$ by the choice of δ. If x is irrational, $f(x) = 0 = f(a)$. If $x = p/q$ is rational in lowest terms, its denominator cannot be at most Q, since that would put x in F. Thus $q > Q$ and

$$|f(x) - f(a)| = \frac{1}{q} < \frac{1}{Q} < \varepsilon.$$

The same strict error bound is immediate in the irrational case. This proves continuity at a by Theorem 17.2.

Discontinuity at a rational point: approach through irrationals.

Fix a rational $a = p/q$ in lowest terms, allowing $0 = 0/1$. Then $f(a) = 1/q > 0$. By irrational density, choose irrational $t_n \in (a - 1/n, a + 1/n)$ for each n. These points tend to a, while $f(t_n) = 0$ for every n. The image limit is zero and differs from the positive value $f(a)$, so f is discontinuous at a.

Conclusion. The continuity set is exactly $\mathbb{R} \setminus \mathbb{Q}$; the discontinuity set is exactly $\mathbb{Q}$.

17.15 Exercise 17.15: A continuity test that omits the point itself

What we must show. Prove that f is continuous at a domain point x_0 if and only if $f(x_n) \to f(x_0)$ for every sequence in $\mathrm{dom}(f) \setminus \{x_0\}$ tending to x_0.

Forward implication: the omitted-point sequences are a subclass.

If f is continuous at x_0, Definition 17.1 gives the image limit for every domain sequence approaching x_0. It therefore gives it for each sequence that avoids x_0.

Reverse implication: split an arbitrary sequence by its equal terms.

Assume the omitted-point condition. Let (y_n) be any domain sequence with $y_n \to x_0$. At indices where $y_n = x_0$, the image error is exactly zero: $|f(y_n) - f(x_0)| = 0$. If only finitely many indices have $y_n \neq x_0$, the sequence is eventually equal to x_0, so its images are eventually equal to $f(x_0)$ and converge to that value.

If infinitely many indices have $y_n \neq x_0$, list all of them in increasing order as $n_1 < n_2 < \cdots$. The subsequence $x_k = y_{n_k}$ avoids x_0 and still tends to x_0. The assumption gives $f(x_k) \to f(x_0)$. Given $\varepsilon > 0$, choose an integer $K \geq 1$ so that $k > K$ implies $|f(x_k) - f(x_0)| < \varepsilon$. For any original index $n > n_K$, either $y_n = x_0$, giving zero error, or $n = n_k$ for some $k > K$, giving the same strict error bound by the selected subsequence. Thus $f(y_n) \to f(x_0)$ for the original full sequence. As the sequence was arbitrary, Definition 17.1 proves continuity.

Check what happens at an isolated domain point.

If some radius $r > 0$ satisfies

$$\mathrm{dom}(f) \cap (x_0 - r, x_0 + r) = \{x_0\},$$

there is no sequence of domain points avoiding x_0 that converges to x_0: convergence would eventually place its points in this ball, where no such points exist. The omitted-point condition is then vacuously true. Every domain sequence converging to x_0 is eventually x_0, so continuity is automatically true as well.

Conclusion. The punctured-sequence condition is equivalent to continuity, including at isolated points of the domain.

17.16 Exercise 17.16: Jumps in the postage-stamp function

What we must show. For fixed postage charges A and B, with the additional ounce charge $B > 0$, prove that the stated function P is discontinuous at every positive integer.

Plan. Approach an integer from the preceding interval. Its charge differs from the value at the integer by exactly B.

Write the domain and the two adjacent charges.

The domain is $[0, \infty)$. The charge is A on $[0, 1)$, and is $A + Bn$ on $[n, n + 1)$ for every positive integer n. Equivalently, the latter formula also covers $n = 0$. Fix any positive integer m. Its value is $P(m) = A + Bm$, because $m \in [m, m + 1)$. On the interval $[m - 1, m)$, the value is instead $A + B(m - 1)$.

Select domain points in the preceding interval.

For each $n \in \mathbb{N}$, set

$$x_n = m - \frac{1}{n+1}.$$

Since $0 < 1/(n + 1) < 1$, we have $m - 1 < x_n < m$, so x_n belongs to the domain and to the preceding charge interval. Also $|x_n - m| = 1/(n + 1) \to 0$, hence $x_n \to m$. But

$$P(x_n) = A + B(m - 1) \longrightarrow A + B(m - 1) \neq A + Bm = P(m).$$

The inequality holds because their difference is the nonzero charge B. Definition 17.1 therefore gives discontinuity at m. Since m was any positive integer, all such integers are discontinuity points.

Record the implicit nonzero-charge assumption.

The proof works for any fixed $B \neq 0$, although postage gives the natural assumption $B > 0$. If $B = 0$, the formula reduces to the constant function $P(x) = A$, which is continuous; the requested jump assertion necessarily presumes a nonzero additional-ounce charge. Charges here are fixed for the rate schedule under consideration.

Conclusion. With $B \neq 0$, each positive integer has a jump of magnitude $|B|$, so P is discontinuous at each of them.

18 Properties of continuous functions

The extreme value theorem concerns a continuous function on a closed, bounded interval. The intermediate value theorem concerns a continuous function on any interval. These conclusions answer different questions: the first guarantees attained extreme values, while the second fills in values between two already attained values.

All continuity statements below are relative to the specified domain. For example, an inverse defined on a half-closed interval is tested only at points belonging to that interval. Multiplying an inequality by a negative number reverses its direction; we will use that fact explicitly.

18.1 Exercise 18.1: A maximum after reflection is a minimum before reflection

What we must show. Suppose f is continuous on $[a, b]$ and $-f$ assumes its maximum at $x_0 \in [a, b]$. Show that f assumes its minimum there.

Translate the meaning of maximum into an inequality.

For every $x \in [a, b]$, the maximum assumption means

$$(-f)(x) \leq (-f)(x_0), \qquad \text{that is,} \qquad -f(x) \leq -f(x_0).$$

Multiplying both sides by -1 reverses the order and gives $f(x) \geq f(x_0)$. Equivalently,

$$f(x_0) \leq f(x) \qquad \text{for every } x \in [a, b].$$

Because x_0 belongs to the domain, the value $f(x_0)$ is actually attained, rather than merely being a lower bound.

Conclusion. The displayed inequality is exactly the definition of a minimum of f at x_0. Continuity is used by the theorem to obtain the maximum of $-f$; this final order reversal needs no additional continuity argument.

18.2 Exercise 18.2: Where the closed-interval proof uses the endpoints

What we must show. Locate the failure in the proof of Theorem 18.1 when its domain $[a, b]$ is replaced by (a, b), and explain the consequences.

1. A bounded input sequence still has a convergent subsequence.

Every sequence (x_n) in (a, b) is bounded, so Bolzano–Weierstrass still provides a subsequence x_{n_k} converging to a real number x_0. The difficulty is its location. The inequalities $a < x_{n_k} < b$ imply only $a \leq x_0 \leq b$. They do not imply the strict inequalities $a < x_0 < b$. Thus the subsequence may tend to an omitted endpoint.

2. Identify the missing step in the boundedness argument.

In the theorem's contradiction argument, the selected inputs satisfy $|f(x_n)| > n$. With a closed interval, the subsequential limit x_0 lies in the domain, so continuity gives $f(x_{n_k}) \to f(x_0)$, a finite real number. On an open interval, x_0 may be outside the domain. Then $f(x_0)$ is not defined, and continuity of f on (a, b) gives no statement at x_0. The contradiction is lost precisely at that step.

For a concrete example, $f(x) = 1/x$ is continuous on $(0, 1)$. The sequence $x_n = 1/(n + 1)$ lies in that interval and tends to the excluded point 0, while $f(x_n) = n + 1$ is unbounded. There is no value $f(0)$ to which continuity could force these outputs to converge.

3. The same missing-domain issue affects attainment of the bounds.

Even if f is bounded, a sequence (y_n) chosen so that $f(y_n) \to \sup f((a, b))$ may have its input subsequence tend to an endpoint. One then cannot identify the limiting output as a value of f at an allowed input.

For example, $f(x) = x$ is continuous and bounded on $(0, 1)$, with infimum 0 and supremum 1. It has neither a minimum nor a maximum there. In fact, for every $x \in (0, 1)$,

$$0 < \frac{x}{2} < x < \frac{x + 1}{2} < 1.$$

The points $x/2$ and $(x + 1)/2$ give strictly smaller and strictly larger function values than $f(x)$. The sequences approaching the bounds tend to the excluded endpoints 0 and 1.

Conclusion. Bolzano–Weierstrass still works, but its limit need not belong to the open domain. Both the boundedness contradiction and the extremum-attainment step require continuity at that limit. An individual function on an open interval may still be bounded and attain extrema; those properties are no longer guaranteed.

18.3 Exercise 18.3: A polynomial on a half-open interval

What we must show. Use calculus to determine the extreme values of

$$f(x) = x^3 - 6x^2 + 9x + 1 \qquad (0 \le x < 5).$$

Plan. Find the stationary points, inspect the derivative's sign, and remember that 5 is not an allowed input.

1. Differentiate and find the interior stationary points.

The power rule gives

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3).$$

Thus the only zeros of f' are 1 and 3, both in the domain. Its signs on the intervening intervals are

x	$(0, 1)$	$(1, 3)$	$(3, 5)$
$x - 1$	$-$	$+$	$+$
$x - 3$	$-$	$-$	$+$
$f'(x)$	$+$	$-$	$+$

The mean value theorem therefore makes f increasing on $[0, 1]$, decreasing on $[1, 3]$, and increasing on $[3, 5)$.

2. Evaluate all the candidates and the omitted endpoint value.

Substitution gives

$$f(0) = 1,$$
$$f(1) = 1 - 6 + 9 + 1 = 5,$$
$$f(3) = 27 - 54 + 27 + 1 = 1,$$
$$f(5) = 125 - 150 + 45 + 1 = 21.$$

The final line evaluates the polynomial formula at 5 to find the endpoint limit. It does not add 5 to the stated domain.

3. Establish and locate the global minimum.

The useful factorization

$$f(x) = x(x-3)^2 + 1$$

shows directly that $f(x) \geq 1$ for every $x \in [0, 5)$: the factor x and the square $(x-3)^2$ are nonnegative. Equality requires $x(x-3)^2 = 0$, which holds exactly when $x = 0$ or $x = 3$. Both points are in the domain.

4. Explain why there is no global maximum.

On $[0, 3]$, the derivative analysis gives $f(x) \leq 5$. On $[3, 5)$, strict increase toward the omitted endpoint gives $f(x) < 21$. In addition, $x_n = 5 - 1/n$ lies in $[0, 5)$ and tends to 5, so continuity of the polynomial formula on $\mathbb{R}$ gives $f(x_n) \to 21$. Thus 21 is the supremum but is not attained.

One can also exclude a maximum without using the word supremum. For $x \in [3, 5)$, the point $y = (x + 5)/2$ satisfies $x < y < 5$, and strict increase gives $f(y) > f(x)$. For $x \in [0, 3]$, we have $f(x) \leq 5$, whereas $f(4) = 64 - 96 + 36 + 1 = 5$ and strict increase on $[3, 5)$ gives $f(9/2) > f(4) = 5$. Thus every allowed input has some allowed input with a larger function value.

Conclusion. The minimum is 1, attained exactly at $x = 0$ and $x = 3$. There is no maximum. The supremum is 21, approached as $x \to 5$ from the left. The value 5 at $x = 1$ is a local maximum, not a global maximum on $[0, 5)$.

18.4 Exercise 18.4: A missing limit point produces an unbounded continuous function

What we must show. Suppose $S \subseteq \mathbb{R}$ contains a sequence (x_n) converging to $x_0 \notin S$. Construct a continuous function on S whose values are unbounded.

1. Choose a function that grows near the excluded point.

Define

$$F(x) = \frac{1}{|x - x_0|} \qquad (x \in S).$$

Every $x \in S$ differs from x_0, so the denominator is positive and $F(x)$ is a well-defined finite real number. The function is a quotient of continuous functions with a nonzero denominator, restricted to S, and hence is continuous there.

2. Verify continuity directly, including the domain restriction.

Fix $y \in S$ and let $d = |y - x_0| > 0$. If $x \in S$ and $|x - y| < d/2$, the triangle inequality yields

$$|x - x_0| \geq |y - x_0| - |x - y| > d/2.$$

The reverse triangle inequality then gives

$$|F(x) - F(y)| = \frac{||y - x_0| - |x - x_0||}{|x - x_0|\,|y - x_0|} \leq \frac{|x - y|}{|x - x_0|\,d} \leq \frac{2\,|x - y|}{d^2}.$$

Given $\varepsilon > 0$, choose

$$\delta = \min\left\{\frac{d}{2}, \frac{\varepsilon d^2}{2}\right\} > 0.$$

For every $x \in S$ with $|x - y| < \delta$, the estimate is strictly less than ε. Thus F is continuous at every $y \in S$, in precisely the domain-relative sense.

3. Use the given sequence to disprove boundedness.

Let $M > 0$. Since $x_n \to x_0$, there is an integer N such that $n > N$ implies $|x_n - x_0| < 1/M$. All these distances are nonzero because $x_n \in S$ and $x_0 \notin S$. Reciprocating the positive inequality gives

$$F(x_n) = \frac{1}{|x_n - x_0|} > M \qquad (n > N).$$

Thus the function takes values larger than every proposed positive bound. In fact, $F(x_n) \to +\infty$.

Conclusion. The function $F(x) = 1/|x - x_0|$ is continuous on S and unbounded there. Its singular point lies outside its domain.

18.5 Exercise 18.5: Intersecting graphs and a fixed point

What we must show. Prove (a) that the stated endpoint inequalities force two continuous graphs to meet, and (b) that Example 1 follows as a special case.

(a), first: subtract the functions.

Assume f and g are continuous on $[a, b]$, with $f(a) \geq g(a)$ and $f(b) \leq g(b)$. Define $h(x) = f(x) - g(x)$ on this interval. The difference of continuous functions is continuous, and

$$h(a) \geq 0, \qquad h(b) \leq 0.$$

If $h(a) = 0$, take $x_0 = a$. If $h(b) = 0$, take $x_0 = b$. Either choice already gives $f(x_0) = g(x_0)$.

(a), second: use the strict sign change in the remaining case.

If neither endpoint value is zero, the weak inequalities become $h(a) > 0$ and $h(b) < 0$. Thus 0 is strictly between those values. The intermediate value theorem gives $x_0 \in (a, b)$ with $h(x_0) = 0$. By the definition of h, this equation is $f(x_0) - g(x_0) = 0$, or $f(x_0) = g(x_0)$. If the interval degenerates to $a = b$, the two given inequalities at that single point already force equality, so that case also is covered.

(b) Compare a map of the unit interval with the identity.

In Example 1, a continuous function f maps $[0, 1]$ into $[0, 1]$. Let $g(x) = x$ on $[0, 1]$, a continuous polynomial. The image restriction gives

$$f(0) \geq 0 = g(0), \qquad f(1) \leq 1 = g(1).$$

Part (a) therefore supplies $x_0 \in [0,1]$ with $f(x_0) = g(x_0) = x_0$. This is the fixed-point conclusion of the example, and it includes possible fixed points at 0 or 1.

Conclusion. The endpoint order reversal forces an intersection. Comparing a continuous map $[0,1] \to [0,1]$ with $g(x) = x$ recovers Example 1.

18.6 Exercise 18.6: A fixed point of cosine inside a specified interval

What we must show. Prove that $x = \cos x$ for at least one $x \in (0, \pi/2)$.

1. Move both sides to one continuous function.

Set $h(x) = x - \cos x$ on $[0, \pi/2]$. The identity and cosine are continuous, so their difference is continuous on this closed interval.

2. Compute opposite nonzero signs at the endpoints.

The familiar trigonometric values give

$$h(0) = 0 - 1 = -1 < 0, \qquad h(\pi/2) = \pi/2 - 0 = \pi/2 > 0.$$

Consequently the target value 0 is strictly between the endpoint values. By the intermediate value theorem there is $x \in (0, \pi/2)$ with $h(x) = 0$.

Conclusion. At this interior point, $h(x) = 0$ means $x - \cos x = 0$, which is the required equation $x = \cos x$.

18.7 Exercise 18.7: An exponential equation with an interior solution

What we must show. Prove that $xe^x = 2$ for some $x \in (0,1)$.

1. Form a continuous function whose zero solves the equation.

Let $h(x) = xe^x - 2$ on $[0,1]$. The identity and exponential are continuous, and the product and subtraction rules make h continuous as well.

2. Check the signs, including the inequality involving e.

At zero, $e^0 = 1$, so $h(0) = 0 \cdot 1 - 2 = -2 < 0$. At one, $h(1) = e - 2$. The identity $e = \sum_{k=0}^{\infty} 1/k!$, stated in Section 16, shows that

$$e \geq 1 + 1 + \frac{1}{2} = \frac{5}{2} > 2,$$

because the omitted terms are nonnegative. Hence $h(1) > 0$.

3. Apply the intermediate value theorem with strict endpoint signs.

There is $x \in (0,1)$ with $h(x) = 0$, since zero lies strictly between $h(0)$ and $h(1)$.

Conclusion. The resulting equation is $xe^x - 2 = 0$, so $xe^x = 2$ at an allowed interior point.

18.8 Exercise 18.8: Opposite signs force a real zero

What we must show. Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$. Prove that f has a zero strictly between a and b.

1. Extract the endpoint information from the product.

A negative product of two real numbers means one factor is strictly positive and the other strictly negative. In particular, neither factor is zero. Also $a \neq b$, since otherwise the product would be the square $f(a)^2 \geq 0$.

Put $\alpha = \min\{a, b\}$ and $\beta = \max\{a, b\}$. Then $\alpha < \beta$, and $f(\alpha)$ and $f(\beta)$ still are the two values of opposite signs. Their order may be either positive then negative, or negative then positive; in both cases zero lies strictly between them.

2. Restrict to the ordered interval and apply the theorem.

The restriction of f to $[\alpha, \beta]$ is continuous. The intermediate value theorem provides $c \in (\alpha, \beta)$ with $f(c) = 0$.

Conclusion. The point c is strictly between the original endpoints, regardless of which of a and b was larger.

18.9 Exercise 18.9: Every polynomial of odd degree has a real root

What we must show. For a polynomial

$$f(x) = a_0 + a_1 x + \cdots + a_m x^m, \qquad m \text{ odd}, \quad a_m \neq 0,$$

prove that $f(c) = 0$ for some real c.

Plan. At inputs of sufficiently large magnitude, the leading odd power dominates all the lower powers. Give an explicit bound rather than assume a limit theorem for functions from the later Section 20.

1. Normalize the leading coefficient without changing the zeros.

Since $a_m \neq 0$, define

$$g(x) = \frac{f(x)}{a_m} = x^m + \sum_{j=0}^{m-1} b_j x^j, \qquad b_j = \frac{a_j}{a_m}.$$

The functions f and g have exactly the same zeros: $g(x) = 0$ if and only if $f(x) = a_m g(x) = 0$. Both are polynomials and hence are continuous on $\mathbb{R}$.

2. Bound the lower-degree part at large positive and negative inputs.

Set

$$C = \sum_{j=0}^{m-1} |b_j| \geq 0, \qquad R = \max\{1, C\} + 1.$$

Thus $R > 1$ and $R > C$. For either $x = R$ or $x = -R$, we have $|x| = R$. Since $0 \leq j \leq m-1$ and $R \geq 1$, $R^j \leq R^{m-1}$. Applying the triangle inequality gives

$$\left| \sum_{j=0}^{m-1} b_j x^j \right| \leq \sum_{j=0}^{m-1} |b_j| \, R^j \leq C R^{m-1}.$$

3. Use the odd parity of m to obtain opposite signs.

At R, the preceding estimate gives

$$g(R) \geq R^m - CR^{m-1} = R^{m-1}(R - C) > 0.$$

At $-R$, the leading term is $(-R)^m = -R^m$ because m is odd, so

$$g(-R) \leq -R^m + CR^{m-1} = -R^{m-1}(R - C) < 0.$$

Both strict signs follow from $R > 0$ and $R - C > 0$. The argument includes $m = 1$: the lower-degree part then is just the constant b_0.

4. Fill in zero between those two values.

Continuity of g on $[-R, R]$ and its opposite endpoint signs give a point $c \in (-R, R)$ with $g(c) = 0$ by the intermediate value theorem. Hence $f(c) = a_m g(c) = 0$.

Conclusion. Every real polynomial of odd degree has a real root. The proof treats every nonzero leading coefficient, including a negative one, through the normalization $g = f/a_m$.

18.10 Exercise 18.10: Equal values exactly one unit apart

What we must show. Suppose f is continuous on $[0, 2]$ and $f(0) = f(2)$. Find $x, y \in [0, 2]$ with $|y - x| = 1$ and $f(x) = f(y)$.

1. Compare the values separated by one unit.

For $t \in [0, 1]$, both t and $t + 1$ belong to the domain $[0, 2]$. Define

$$h(t) = f(t + 1) - f(t).$$

Translation $t \mapsto t + 1$ is continuous, so composition makes $t \mapsto f(t + 1)$ continuous on $[0, 1]$. The other term is a restriction of f, so h is continuous.

2. Show that its endpoint values are negatives of one another.

The given equality $f(2) = f(0)$ gives

$$\begin{aligned} h(0) &= f(1) - f(0), \\ h(1) &= f(2) - f(1) = f(0) - f(1) = -h(0). \end{aligned}$$

If $h(0) = 0$, choose $t_0 = 0$. If $h(0) \neq 0$, these two endpoint values have opposite signs. The intermediate value theorem then supplies $t_0 \in (0, 1)$ with $h(t_0) = 0$. Thus in every case there is $t_0 \in [0, 1]$ with $h(t_0) = 0$.

3. Translate the zero back into the requested pair.

Take $x = t_0$ and $y = t_0 + 1$. Then $x \in [0, 1]$ and $y \in [1, 2]$, so both belong to $[0, 2]$. Moreover,

$$|y - x| = |(t_0 + 1) - t_0| = 1, \qquad f(y) - f(x) = h(t_0) = 0.$$

Conclusion. The constructed pair has precisely the required distance and equal function values.

18.11 Exercise 18.11: The decreasing versions of the inverse-function theorems

What we must show. Prove (a) that negating a strictly increasing function makes it strictly decreasing, and (b) state and prove the strictly decreasing versions of Theorems 18.4 and 18.5.

(a) Negation reverses each strict inequality.

If f is strictly increasing on I, then $x_1 < x_2$ in I implies $f(x_1) < f(x_2)$. Multiplication by -1 gives $-f(x_1) > -f(x_2)$. This is exactly the definition of strict decrease of $-f$ on I. The same reasoning shows that negating a strictly decreasing function makes it strictly increasing.

(b), the decreasing version of Theorem 18.5.

Statement. If g is strictly decreasing on an interval J and its image $g(J) = I$ is an interval, then g is continuous on J.

Proof. Let $h = -g$. Part (a), with the inequalities reversed, makes h strictly increasing. Its image is

$$h(J) = -I = \{-y : y \in I\}.$$

Reflection preserves the interval property. Indeed, if $u < v$ belong to $-I$ and $u < w < v$, then $-v < -w < -u$. The endpoints $-v, -u$ belong to I, so the interval property gives $-w \in I$, or $w \in -I$. Thus $-I$ is an interval, with any included endpoints and any infinite endpoints reflected in the usual way.

The increasing version of Theorem 18.5 applies to h: its domain is the interval J, it is strictly increasing, and its image $-I$ is an interval. Therefore h is continuous. Negation is continuous, so $g = -h$ is continuous by Theorem 17.3. Explicitly, the relevant errors are equal:

$$|g(x) - g(x_0)| = |-h(x) + h(x_0)| = |h(x) - h(x_0)|\,.$$

Thus the very same δ that works for h at x_0 works for g. This proves the decreasing version of Theorem 18.5.

(b), the decreasing version of Theorem 18.4.

Statement. If f is continuous and strictly decreasing on an interval I, then its image $J = f(I)$ is an interval, its inverse is a function $f^{-1} : J \to I$, and that inverse is continuous and strictly decreasing on J.

Proof. The intermediate value theorem and Corollary 18.3 make $J = f(I)$ an interval. Strict decrease makes f one-to-one: distinct inputs can be ordered, and that order gives distinct outputs. Thus every $u \in J$ has exactly one preimage in I, which defines $f^{-1}(u)$.

Take $u < v$ in J, and set $x = f^{-1}(u)$ and $y = f^{-1}(v)$. If $x = y$, then $u = f(x) = f(y) = v$, a contradiction. If $x < y$, strict decrease gives $u = f(x) > f(y) = v$, also a contradiction. Hence $x > y$, which means $f^{-1}(u) > f^{-1}(v)$. The inverse is strictly decreasing.

Its domain J is an interval and its image is exactly I, also an interval. The decreasing version of Theorem 18.5 just proved therefore makes f^{-1} continuous on J.

No openness of either image interval has been assumed. Continuity at an included endpoint is relative to J; the definition tests only $u \in J$. If the domain is a singleton, strict monotonicity is vacuous and the singleton inverse is continuous directly, so the statements still hold.

Conclusion. Both decreasing theorems have the same conclusions as their increasing counterparts, with strict decrease replacing strict increase. Reflection and the interval-image condition supply the continuity argument.

18.12 Exercise 18.12: The intermediate value property without continuity

What we must show. For

$$f(x) = \begin{cases} \sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases}$$

show (a) discontinuity at zero and (b) the intermediate value property on all of $\mathbb{R}$.

(a) Give an input sequence whose outputs do not approach $f(0)$.

Let

$$u_n = \frac{1}{2\pi n + \pi/2} \qquad (n \in \mathbb{N}).$$

The positive denominators grow without bound, so $u_n \to 0$. But periodicity of sine gives

$$f(u_n) = \sin(2\pi n + \pi/2) = 1 \qquad \text{for every } n,$$

whereas $f(0) = 0$. Thus $f(u_n)$ does not tend to $f(0)$, contradicting the sequential condition for continuity. This is the same discontinuity established in Exercise 17.10(b).

(b), preparation: state exactly what must be checked.

Fix any $a < b$, and let y lie strictly between $f(a)$ and $f(b)$. Every value of f belongs to $[-1, 1]$, so the strict intermediate value y satisfies $-1 < y < 1$. We must find $x \in (a, b)$ with $f(x) = y$.

(b), intervals that stay away from zero.

If $0 \notin [a, b]$, the reciprocal function $1/x$ is continuous on $[a, b]$, and composition with sine makes f continuous there. The ordinary intermediate value theorem then supplies the required interior point.

(b), intervals containing zero and some positive points.

Suppose $0 \in [a, b]$ and $b > 0$. For $n \in \mathbb{N}$, put

$$u_n = \frac{1}{2\pi n + \pi/2}, \qquad v_n = \frac{1}{2\pi n + 3\pi/2}.$$

Both sequences tend to zero, and $0 < v_n < u_n$. Choose n large enough that $u_n < b$. Since $a \leq 0$, the entire interval $[v_n, u_n]$ lies in (a, b) and excludes zero. On its endpoints,

$$f(v_n) = \sin(2\pi n + 3\pi/2) = -1, \qquad f(u_n) = 1.$$

The function is continuous on $[v_n, u_n]$ by the composition argument. Since $-1 < y < 1$, the intermediate value theorem gives $x \in (v_n, u_n) \subseteq (a, b)$ with $f(x) = y$.

(b), the remaining intervals end at zero from the left.

If $0 \in [a, b]$ but $b \leq 0$, then $b = 0$ and $a < 0$. Choose n large enough that $u_n < -a$. Then

$$a < -u_n < -v_n < 0 = b.$$

For nonzero inputs f is odd, because $f(-t) = \sin(-1/t) = -\sin(1/t) = -f(t)$. Therefore $f(-u_n) = -1$ and $f(-v_n) = 1$. The function is continuous on the zero-free interval $[-u_n, -v_n]$. The intermediate value theorem again gives an interior point with value y, and that point lies in (a, b).

These cases cover every possible position of $[a, b]$. When the endpoint values are equal there is no strictly intermediate y to check, so that case imposes no additional condition.

Conclusion. The function is discontinuous at zero but has the intermediate value property on $\mathbb{R}$. Continuity implies that property, while the property alone does not imply continuity.

19 Uniform continuity

Uniform continuity on a set S means that, for each $\varepsilon > 0$, one positive δ works for *every* pair $x, y \in S$:

$$|x - y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon.$$

The choice may depend on ε and on the fixed set S, but not on the chosen points. Restricting a uniformly continuous function to a smaller set preserves this implication, since it reduces the number of pairs to check.

We use Theorem 19.2 for continuous functions on closed bounded intervals and Theorem 19.6 for continuous interval functions with bounded derivative. A bound $|f'| \leq M$ with $M > 0$ gives $|f(x) - f(y)| \leq M\,|x - y|$ by the Mean Value theorem, so $\delta = \varepsilon/M$ works. Derivatives are used only where the stated interval and differentiability hypotheses are satisfied.

19.1 Exercise 19.1: Seven tests for uniform continuity

What we must show. Decide uniform continuity on each specified domain, and justify each answer. In part (e) the function is $1/x^3$, and in part (f) the sine argument is $1/x^2$.

(a) The continuous expression on $[0, \pi]$.

For

$$f(x) = x^{17} \sin x - e^x \cos(3x),$$

the power, sine, exponential, and cosine functions are continuous. Composition gives continuity of $\cos(3x)$, multiplication gives continuity of both products, and subtraction gives continuity of f on $\mathbb{R}$. In particular it is continuous on the closed bounded interval $[0, \pi]$. Theorem 19.2 makes f uniformly continuous there. No estimate involving its complicated derivative is needed.

(b) The cube on $[0, 1]$.

The polynomial x^3 is continuous on $\mathbb{R}$, hence on $[0, 1]$. Theorem 19.2 therefore proves uniform continuity on $[0, 1]$. One can also see a point-independent estimate directly: if $0 \leq x, y \leq 1$, then

$$\left|x^3 - y^3\right| = |x - y| \left|x^2 + xy + y^2\right| \leq 3\,|x - y|\,.$$

Thus $\delta = \varepsilon/3$ works.

(c) The cube on $(0, 1)$.

The function in (c) is the restriction of the uniformly continuous function in (b). The same δ therefore works for all pairs in $(0, 1)$. A domain need not be closed for its function to be uniformly continuous.

(d) The cube on all of $\mathbb{R}$.

For $n \in \mathbb{N}$, choose

$$x_n = n, \qquad y_n = n + \frac{1}{n^2}.$$

Their distance $|x_n - y_n| = 1/n^2$ tends to zero, but expansion of the cube gives

$$y_n^3 - x_n^3 = 3n^2 \frac{1}{n^2} + 3n \frac{1}{n^4} + \frac{1}{n^6} = 3 + \frac{3}{n^3} + \frac{1}{n^6} \geq 3.$$

Fix $\varepsilon_0 = 1$. For any proposed $\delta > 0$, choose n so large that $1/n^2 < \delta$. The corresponding pair is closer than δ, but its function values differ by at least $3 > \varepsilon_0$. This violates Definition 19.1. The cube is not uniformly continuous on $\mathbb{R}$.

(e) The reciprocal cube on $(0, 1]$.

The sequence $x_n = 1/n$ lies in $(0, 1]$ and converges to zero, so it is a Cauchy sequence. For $f(x) = 1/x^3$ its image is $f(x_n) = n^3$, which is unbounded. Every real Cauchy sequence is bounded, so this image is not a Cauchy sequence. If f were uniformly continuous, Theorem 19.4 would make the image of every Cauchy sequence a Cauchy sequence. This contradiction proves that $1/x^3$ is not uniformly continuous on $(0, 1]$.

(f) The rapidly oscillating sine on $(0, 1]$.

Let $f(x) = \sin(1/x^2)$ and choose

$$x_n = \frac{1}{\sqrt{2\pi n + \pi/2}}, \qquad y_n = \frac{1}{\sqrt{2\pi n + 3\pi/2}}.$$

The denominators exceed 1, so both sequences lie in $(0, 1]$. Both tend to zero, whence $|x_n - y_n| \leq x_n + y_n \to 0$. On the other hand,

$$f(x_n) = \sin(2\pi n + \pi/2) = 1, \qquad f(y_n) = \sin(2\pi n + 3\pi/2) = -1.$$

Their images always differ by 2. Taking $\varepsilon_0 = 1$, every proposed $\delta > 0$ is defeated by a sufficiently late pair with $|x_n - y_n| < \delta$. Thus f is not uniformly continuous on this interval.

(g) The damped oscillation on $(0, 1]$.

For $x > 0$ let $f(x) = x^2 \sin(1/x)$, and extend the formula by $F(0) = 0$, $F(x) = f(x)$ for $0 < x \leq 1$. For positive x it is a product and composition of continuous functions. At zero,

$$|F(x) - F(0)| = \left|x^2 \sin(1/x)\right| \leq x^2.$$

Given $\varepsilon > 0$, the choice $\delta = \sqrt{\varepsilon}$ makes this error less than ε whenever $0 \leq x < \delta$. Thus F is continuous on $[0, 1]$. Theorem 19.2 makes F uniformly continuous there, and restriction proves uniform continuity of f on $(0, 1]$.

Conclusion. Parts (a), (b), (c), and (g) are uniformly continuous on their specified sets. Parts (d), (e), and (f) are not.

19.2 Exercise 19.2: Three direct epsilon–delta proofs

What we must show. Verify Definition 19.1 directly for each function, choosing δ independently of the pair of points.

(a) The affine function $3x + 11$ on $\mathbb{R}$.

Let $\varepsilon > 0$ and set $\delta = \varepsilon/3 > 0$. For arbitrary $x, y \in \mathbb{R}$ with $|x - y| < \delta$, cancellation of the constants gives

$$|(3x + 11) - (3y + 11)| = 3\,|x - y| < 3\delta = \varepsilon.$$

This one δ works throughout the unbounded domain.

(b) The square on $[0,3]$.

For $x, y \in [0,3]$, one has $0 \le x+y \le 6$. Factor the difference of squares to get

$$\left|x^2 - y^2\right| = |x-y|\,|x+y| \le 6\,|x-y|\,.$$

For any $\varepsilon > 0$, take $\delta = \varepsilon/6$. Then $|x-y| < \delta$ makes the displayed error less than $6\delta = \varepsilon$, for every such pair.

(c) The reciprocal on $[1/2, +\infty)$.

If $x, y \ge 1/2$, both denominators are positive and $xy \ge 1/4$. Consequently

$$\left|\frac{1}{x} - \frac{1}{y}\right| = \frac{|y-x|}{xy} \le 4\,|x-y|\,.$$

Choose $\delta = \varepsilon/4$. For every pair in the domain with distance less than δ, the function-value distance is less than $4\delta = \varepsilon$.

Conclusion. The respective choices $\varepsilon/3$, $\varepsilon/6$, and $\varepsilon/4$ establish all three uniform-continuity claims directly from the definition.

19.3 Exercise 19.3: Direct proofs for two rational functions

What we must show. Repeat the direct epsilon–delta verification for $x/(x+1)$ on $[0,2]$ and $5x/(2x-1)$ on $[1,+\infty)$.

(a) Subtract with the denominator kept explicit.

For $x, y \in [0,2]$, both $x+1$ and $y+1$ are at least 1. Common denominators and cancellation give

$$\begin{aligned}\left|\frac{x}{x+1} - \frac{y}{y+1}\right| &= \frac{|x(y+1) - y(x+1)|}{(x+1)(y+1)} \\ &= \frac{|xy + x - xy - y|}{(x+1)(y+1)} = \frac{|x-y|}{(x+1)(y+1)} \le |x-y|\,.\end{aligned}$$

Given $\varepsilon > 0$, set $\delta = \varepsilon$. The final bound then becomes less than ε for every pair with $|x-y| < \delta$.

(b) Do the same calculation on the unbounded interval.

For $x, y \ge 1$, one has $2x - 1 \ge 1$ and $2y - 1 \ge 1$. Thus

$$\begin{aligned}\left|\frac{5x}{2x-1} - \frac{5y}{2y-1}\right| &= \frac{|5x(2y-1) - 5y(2x-1)|}{(2x-1)(2y-1)} \\ &= \frac{5\,|-x+y|}{(2x-1)(2y-1)} \le 5\,|x-y|\,.\end{aligned}$$

Set $\delta = \varepsilon/5$. The error is then less than $5\delta = \varepsilon$ whenever $|x-y| < \delta$.

Conclusion. Both functions are uniformly continuous. The direct choices are $\delta = \varepsilon$ in (a) and $\delta = \varepsilon/5$ in (b).

19.4 Exercise 19.4: A uniformly continuous function on a bounded set is bounded

What we must show. Prove the boundedness assertion in (a), without assuming the domain is an interval or is closed, and apply it to part (b).

(a), preparation: account for the empty domain.

If $S = \varnothing$, every bound is a bound on its function values, so the conclusion is immediate. Suppose henceforth that S is nonempty and bounded, and that f is uniformly continuous on S.

(a), contradiction: select values that are too large.

If f were unbounded, then for every integer $n \geq 1$ some $x_n \in S$ would satisfy $|f(x_n)| > n$. Otherwise a bound on all $|f(x)|$ would exist, which is exactly boundedness of f. Since S is bounded, the real sequence (x_n) is bounded. Bolzano–Weierstrass (Theorem 11.5) supplies a subsequence x_{n_j} converging to some real number x_0. It is not necessary that x_0 belong to S.

Every convergent real sequence is Cauchy, so (x_{n_j}) is a Cauchy sequence of points in S. Theorem 19.4 then makes $(f(x_{n_j}))$ a Cauchy sequence, hence a bounded sequence. However, its construction gives

$$\left|f(x_{n_j})\right| > n_j \geq j.$$

Here $n_j \geq j$ because the selected indices are strictly increasing positive integers. The displayed lower bound makes the image sequence unbounded, a contradiction. Thus f is bounded.

(b) Apply the assertion to the reciprocal square.

The set $(0, 1)$ is bounded. For $n \geq 1$ the points $x_n = 1/(n + 1)$ lie in $(0, 1)$, while

$$\frac{1}{x_n^2} = (n + 1)^2 \longrightarrow +\infty.$$

The function $1/x^2$ is therefore unbounded on this bounded domain. If it were uniformly continuous, part (a) would make it bounded. Hence it is not uniformly continuous on $(0, 1)$.

Conclusion. Uniform continuity implies boundedness on every bounded real domain, including domains with missing limit points. This excludes uniform continuity of $1/x^2$ on $(0, 1)$.

19.5 Exercise 19.5: Six domains for trigonometric and reciprocal functions

What we must show. Determine which of the six specified restrictions are uniformly continuous, with the domain included in each argument.

(a) The tangent on $[0, \pi/4]$.

Cosine has no zero in this closed interval. Thus $\tan x = \sin x / \cos x$ is continuous there, and Theorem 19.2 gives uniform continuity.

(b) The tangent on $[0, \pi/2)$.

This domain is bounded, but tangent is unbounded on it. For example, let $x_n = \pi/2 - 1/n$, which lies in the domain. Then

$$\sin x_n \to 1, \qquad \cos x_n \to 0, \qquad \cos x_n > 0.$$

For any $M > 0$, the first convergence gives $\sin x_n > 1/2$ eventually, and the second gives $0 < \cos x_n < 1/(2M)$ eventually. At indices satisfying both estimates, $\tan x_n > M$. Thus tangent is unbounded and cannot be uniformly continuous on this bounded set by Exercise 19.4(a).

(c) The function $(\sin^2 x)/x$ on $(0, \pi]$.

We need the elementary estimate $|\sin t| \le |t|$. It follows from the Mean Value theorem applied to sine between 0 and t: its derivative is cosine, with absolute value at most 1. Exercise 19.8 supplies the more general version. For $x > 0$ this gives

$$0 \le \frac{\sin^2 x}{x} \le \frac{x^2}{x} = x.$$

Extend the function to $[0, \pi]$ by assigning value 0 at zero. The displayed bound proves continuity at zero: if $0 \le x < \varepsilon$, its value has absolute value less than ε. Away from zero the quotient is continuous. The extension is therefore continuous on $[0, \pi]$, hence uniformly continuous by Theorem 19.2. Its restriction to $(0, \pi]$ is uniformly continuous.

(d) The function $1/(x-3)$ on $(0, 3)$.

For $x_n = 3 - 1/n$, one has $2 \le x_n < 3$, so x_n is in the domain, and

$$\frac{1}{x_n - 3} = -n.$$

The function is unbounded on the bounded set $(0, 3)$. Exercise 19.4(a) therefore excludes uniform continuity.

(e) The same function on $(3, +\infty)$.

This domain is unbounded, so unboundedness of the function alone would not settle uniform continuity. Instead take $x_n = 3 + 1/n$. This converges to 3 and is therefore a Cauchy sequence of points in the domain. Its image is

$$\frac{1}{x_n - 3} = n,$$

which is unbounded and hence is not Cauchy. The Cauchy-sequence preservation theorem (Theorem 19.4) excludes uniform continuity.

(f) The same function on $(4, +\infty)$.

For $x, y > 4$, both $x - 3$ and $y - 3$ exceed 1. Thus

$$\left| \frac{1}{x-3} - \frac{1}{y-3} \right| = \frac{|x-y|}{(x-3)(y-3)} \le |x-y|.$$

The choice $\delta = \varepsilon$ works for every pair in this domain. Hence the function is uniformly continuous here.

Conclusion. Parts (a), (c), and (f) are uniformly continuous; parts (b), (d), and (e) are not.

19.6 Exercise 19.6: Uniform continuity without a bounded derivative

What we must show. For $f(x) = \sqrt{x}$, show the derivative is unbounded on $(0, 1]$ while f is uniformly continuous there, and prove uniform continuity on $[1, +\infty)$.

(a), derivative: exhibit arbitrarily large values.

For $x > 0$, elementary differentiation gives $f'(x) = 1/(2\sqrt{x})$. The points $x_n = 1/n^2$ belong to $(0, 1]$ and satisfy

$$f'(x_n) = \frac{1}{2(1/n)} = \frac{n}{2} \longrightarrow +\infty.$$

Thus the derivative is unbounded on this interval.

(a), the function: use the square-root difference estimate.

For $x \geq y \geq 0$, both $\sqrt{x} - \sqrt{y}$ and $\sqrt{x} + \sqrt{y}$ are nonnegative, and the latter is at least the former. Therefore

$$x - y = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq (\sqrt{x} - \sqrt{y})^2.$$

Taking nonnegative square roots yields $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. If $y > x$, interchange the two points to obtain the same estimate. For $\varepsilon > 0$, choose $\delta = \varepsilon^2$. Whenever $x, y \in (0, 1]$ and $|x - y| < \delta$, it follows that

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|} < \sqrt{\delta} = \varepsilon.$$

This establishes uniform continuity despite the unbounded derivative. Theorem 19.6 gives a sufficient condition, not a necessary condition, so there is no contradiction.

(b) Rationalize on the domain bounded away from zero.

For $x, y \geq 1$, one has $\sqrt{x} + \sqrt{y} \geq 2$. Hence

$$|\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \leq \frac{1}{2} |x - y| .$$

Taking $\delta = 2\varepsilon$ proves uniform continuity on $[1, +\infty)$ directly.

Conclusion. The square root is uniformly continuous on both specified domains. Its derivative is nevertheless unbounded near zero.

19.7 Exercise 19.7: Joining a compact initial interval to a uniform tail

What we must show. Prove part (a)'s extension from a tail to the whole nonnegative half-line, and apply it to square root in part (b).

(a), preparation: choose overlapping intervals.

Let f be continuous on $[0, +\infty)$ and uniformly continuous on $[k, +\infty)$ as in the hypothesis. We only need the part of that tail contained in its domain. Put $K = \max\{k, 0\}$; restriction gives uniform continuity on $[K, +\infty)$. If $k \leq 0$ and the stated tail hypothesis is meaningful, it already includes $[0, +\infty)$, but the argument below also works.

The restriction to $[0, K + 1]$ is continuous on a closed bounded interval, hence uniformly continuous by Theorem 19.2. These two intervals overlap on $[K, K + 1]$. The overlap will handle pairs that are close but lie on opposite sides of the cutoff.

(a), verification: use one delta for every pair.

Fix $\varepsilon > 0$. Choose $\delta_0 > 0$ for uniform continuity on $[0, K+1]$, and $\delta_1 > 0$ for uniform continuity on $[K, +\infty)$, both for this same tolerance ε. Set

$$\delta = \min\{\delta_0, \delta_1, 1\} > 0.$$

Take $x, y \geq 0$ with $|x - y| < \delta$. Interchanging them if necessary, assume $x \leq y$. If $y \leq K + 1$, both points lie in the first interval and their distance is less than δ_0, so their function values differ by less than ε. If $y > K + 1$, then $y - x < 1$, which rearranges to $x > y - 1 > K$. Thus both points lie in the tail, and their distance is less than δ_1, so the same conclusion holds. These two cases exhaust every pair. Thus f is uniformly continuous on the whole half-line.

(b) Apply part (a) to square root.

The function $\sqrt{x}$ is continuous on $[0, +\infty)$ by Exercise 17.4. Exercise 19.6(b) gives uniform continuity on $[1, +\infty)$. Part (a), with $k = 1$, now proves uniform continuity on $[0, +\infty)$.

Conclusion. A continuous function on the half-line is uniformly continuous whenever it is uniformly continuous on a tail. The square root is one such function.

19.8 Exercise 19.8: The sine difference estimate

What we must show. Use the Mean Value theorem to prove $|\sin x - \sin y| \leq |x - y|$, then deduce uniform continuity of sine on $\mathbb{R}$.

(a) Apply the theorem on the interval between the points.

If $x = y$, both sides are zero. Suppose $x < y$. Sine is continuous on $[x, y]$ and differentiable on (x, y), so the Mean Value theorem gives some $c \in (x, y)$ with

$$\sin y - \sin x = (\cos c)(y - x).$$

Taking absolute values and using $|\cos c| \leq 1$ gives

$$|\sin y - \sin x| = |\cos c|\,|y - x| \leq |y - x|\,.$$

If $x > y$, interchange the points; the absolute values make the desired statement unchanged. This proves the estimate for all pairs of real numbers.

(b) Read off a uniform choice.

Given $\varepsilon > 0$, let $\delta = \varepsilon$. For all $x, y \in \mathbb{R}$ with $|x - y| < \delta$, part (a) gives

$$|\sin x - \sin y| \leq |x - y| < \delta = \varepsilon.$$

The choice is independent of both points.

Conclusion. Sine is uniformly continuous on $\mathbb{R}$, with the global difference bound claimed in (a). In particular, $|\sin t| \leq |t|$ follows by setting $y = 0$.

19.9 Exercise 19.9: A damped oscillation on the whole real line

What we must show. For $f(0) = 0$ and $f(x) = x \sin(1/x)$ when $x \neq 0$, verify continuity, uniform continuity on every bounded subset, and decide uniform continuity on $\mathbb{R}$.

(a) Check the origin as well as the ordinary formula.

At every nonzero point, the reciprocal, sine composition, and product are continuous. At zero,

$$|f(x) - f(0)| \le |x|$$

for $x \neq 0$, since $|\sin(1/x)| \le 1$; the inequality also holds at zero. Thus $\delta = \varepsilon$ proves continuity there. The function is continuous on all of $\mathbb{R}$.

(b) Enclose any bounded set in a closed interval.

Let $S \subseteq \mathbb{R}$ be bounded. Choose $R > 0$ so that $S \subseteq [-R, R]$. The function is continuous on this closed bounded interval, hence uniformly continuous there by Theorem 19.2. Restriction gives uniform continuity on S, even if S is not closed or is empty.

(c), tails: bound the derivative away from the origin.

For $x \neq 0$, the product and chain rules give

$$f'(x) = \sin(1/x) - \frac{\cos(1/x)}{x}.$$

If $|x| \ge 1$, then

$$|f'(x)| \le 1 + \frac{1}{|x|} \le 2.$$

Theorem 19.6 therefore gives uniform continuity on each of the intervals $[1, +\infty)$ and $(-\infty, -1]$. More explicitly, the Mean Value theorem gives the bound $|f(x) - f(y)| \le 2\,|x - y|$ whenever both points lie in either one of these intervals.

(c), combine the tails and the central interval.

The central restriction to $[-2, 2]$ is uniformly continuous by part (b). For a fixed $\varepsilon > 0$, choose its uniform parameter $\delta_0 > 0$ and put

$$\delta = \min\{\delta_0, \varepsilon/2, 1\} > 0.$$

Take $x \le y$ with $y - x < \delta$. If both points lie in $[-2, 2]$, the central estimate gives $|f(x) - f(y)| < \varepsilon$. If $y > 2$, then $x > y - 1 > 1$, so both lie in the positive tail; the derivative bound gives $|f(x) - f(y)| \le 2\,|x - y| < 2\delta \le \varepsilon$. If $x < -2$, then $y < x + 1 < -1$, so both lie in the negative tail, and the same bound applies. These cases cover every pair: a pair outside the central square has either $y > 2$ or $x < -2$. Thus the function is uniformly continuous on $\mathbb{R}$.

This last argument also proves a useful joining fact: a continuous real function that is uniformly continuous on both tails $[1, +\infty)$ and $(-\infty, -1]$ is uniformly continuous on $\mathbb{R}$. In the general case, replace $\varepsilon/2$ above by the minimum of the two tail parameters. The same interval classification is unchanged.

Conclusion. The function is continuous on $\mathbb{R}$, uniformly continuous on every bounded subset, and also uniformly continuous on all of $\mathbb{R}$.

19.10 Exercise 19.10: The quadratic damping has uniform control as well

What we must show. Repeat all three parts of Exercise 19.9 for $g(0) = 0$ and $g(x) = x^2 \sin(1/x)$ when $x \neq 0$.

(a) Verify continuity everywhere.

Away from zero, compositions and products give continuity. At zero,

$$|g(x) - g(0)| \leq x^2.$$

For $\varepsilon > 0$, choose $\delta = \sqrt{\varepsilon}$. Then $|x| < \delta$ makes the displayed error less than ε, so g is continuous at zero as well.

(b) Restrict a continuous function on a larger compact interval.

For any bounded $S \subseteq \mathbb{R}$, choose $R > 0$ containing it in $[-R, R]$. Continuity on that interval implies uniform continuity by Theorem 19.2. Restriction proves uniform continuity on S.

(c) Obtain a derivative bound on all of $\mathbb{R}$.

For $x \neq 0$, differentiation gives

$$g'(x) = 2x \sin(1/x) - \cos(1/x).$$

By Exercise 19.8(a), $|\sin(1/x)| \leq 1/|x|$. Thus $|x \sin(1/x)| \leq 1$ for every $x \neq 0$, and

$$|g'(x)| \leq 2\,|x \sin(1/x)| + |\cos(1/x)| \leq 2 + 1 = 3.$$

The derivative at zero exists too. Indeed, for $h \neq 0$,

$$\left| \frac{g(h) - g(0)}{h} - 0 \right| = |h \sin(1/h)| \leq |h| \longrightarrow 0.$$

This is precisely the difference quotient defining $g'(0)$, so $g'(0) = 0$. Consequently g is differentiable on all of $\mathbb{R}$ and satisfies $|g'| \leq 3$ everywhere.

The Mean Value theorem now gives $|g(x) - g(y)| \leq 3\,|x - y|$ for every pair of real points. The choice $\delta = \varepsilon/3$ proves uniform continuity on $\mathbb{R}$, also as an application of Theorem 19.6.

Conclusion. All three conclusions are affirmative: g is continuous on $\mathbb{R}$, uniformly continuous on every bounded subset, and uniformly continuous on $\mathbb{R}$. Its global derivative bound is 3.

19.11 Exercise 19.11: Uniform continuity of the extended sinc function

What we must show. Accept the continuity on $\mathbb{R}$ of the extension

$$\widetilde{h}(x) = \begin{cases} \sin x/x, & x \neq 0, \\ 1, & x = 0, \end{cases}$$

from Example 9, and prove uniform continuity on $\mathbb{R}$.

1. Bound the derivative on both tails.

For $x \neq 0$, the quotient rule gives

$$\widetilde{h}'(x) = \frac{x \cos x - \sin x}{x^2}.$$

If $|x| \geq 1$, the real triangle inequality yields

$$\left|\widetilde{h}'(x)\right| \leq \frac{|x|\,|\cos x| + |\sin x|}{x^2} \leq \frac{1}{|x|} + \frac{1}{x^2} \leq 2.$$

The function is continuous on each tail and differentiable in its interior, so Theorem 19.6 gives uniform continuity on $[1, +\infty)$ and on $(-\infty, -1]$.

2. Join the two tails through a compact middle interval.

The continuity assumed in the exercise gives uniform continuity on $[-2, 2]$ by Theorem 19.2. Fix $\varepsilon > 0$, let $\delta_0 > 0$ be its central parameter, and set $\delta = \min\{\delta_0, \varepsilon/2, 1\}$. For a pair $x \le y$ with $y - x < \delta$, if both are in $[-2, 2]$, the central estimate works. If $y > 2$, then $x > y - 1 > 1$ and both are in the positive tail. If $x < -2$, then $y < x + 1 < -1$ and both are in the negative tail. On either tail the Mean Value theorem and the bound 2 give an error at most $2\,|x - y| < \varepsilon$. Every pair belongs to one of these cases, which proves the uniform-continuity definition on all of $\mathbb{R}$.

Conclusion. The continuous extension of $\sin x/x$ at zero is uniformly continuous on the whole real line. The argument uses the stipulated continuity at zero and derivative bounds only away from zero.

19.12 Exercise 19.12: The running maximum of a continuous function

What we must show. For a continuous $f : [a, b] \to \mathbb{R}$, prove that

$$f^*(x) = \sup\{f(t) : a \le t \le x\}, \qquad x \in [a, b],$$

is increasing and continuous. Increasing here means nondecreasing, as elsewhere in the text.

1. Verify that every supremum is finite and attained.

For each $x \in [a, b]$, the interval $[a, x]$ is nonempty, closed, and bounded. The restriction of f is continuous on it. The extreme value theorem (Theorem 18.1) therefore supplies some $z \in [a, x]$ with $f(z) = \max_{a \le t \le x} f(t)$. In particular the supremum defining $f^*(x)$ is a real number and is this maximum. When $x = a$, the interval is the singleton $\{a\}$ and the assertion reduces to $f^*(a) = f(a)$.

2. Check the order directly from set inclusion.

If $a \le x \le y \le b$, then $[a, x] \subseteq [a, y]$. Every function value used in the supremum for x is therefore also used for y. Hence $f^*(x) \le f^*(y)$. This proves the required increasing property; it need not be strict, as a constant f demonstrates.

3. Obtain one parameter from uniform continuity of f.

Theorem 19.2 makes f uniformly continuous on $[a, b]$. For $\varepsilon > 0$, choose $\delta > 0$ such that

$$u, v \in [a, b], \quad |u - v| < \delta \quad \Longrightarrow \quad |f(u) - f(v)| < \varepsilon.$$

We prove that this same δ works for f^*.

4. Locate a point attaining the larger running maximum.

Take $x, y \in [a, b]$ with $|x - y| < \delta$. If $x = y$, the two running maxima agree. Otherwise interchange the points if needed so that $x < y$. By Step 1 choose $z \in [a, y]$ with $f(z) = f^*(y)$.

If $z \le x$, then $f(z) \le f^*(x)$, since z is among the points considered in that smaller supremum. Combining this with Step 2 gives $f^*(y) \le f^*(x) \le f^*(y)$, so the two values agree and their difference is zero.

If $z > x$, then $0 < z - x \leq y - x < \delta$. The chosen parameter therefore gives $|f(z) - f(x)| < \varepsilon$. Since $f(x) \leq f^*(x)$ and $f(z) = f^*(y)$, we get

$$0 \leq f^*(y) - f^*(x) = f(z) - f^*(x) \leq f(z) - f(x) \leq |f(z) - f(x)| < \varepsilon.$$

The initial inequality again uses monotonicity from Step 2. These two cases prove $|f^*(y) - f^*(x)| < \varepsilon$ for every eligible pair.

Thus f^* is actually uniformly continuous on $[a, b]$. In particular, fixing either point as an arbitrary x_0 verifies continuity at x_0, including the endpoints. If $a = b$, there is only one point and the same assertion is immediate.

Conclusion. The running supremum is a finite attained maximum, is nondecreasing, and is continuous on $[a, b]$; indeed, it is uniformly continuous there.

20 Limits of functions

The approach set is part of a limit's meaning. A two-sided limit at a uses nearby points different from a; a right-hand limit uses points strictly greater than a. The value at a is irrelevant to these punctured limits. Limits may be finite or may equal one of the two infinite symbols. "Does not exist" means that none of these possibilities is a limit.

The sketches below are original schematic plots in finite windows. Open circles mark missing points. Dashed lines mark asymptotes; the axes themselves also mark an asymptote when appropriate. No finite drawing can show all the oscillations accumulating at zero. The formulas and estimates specify those oscillations exactly.

20.1 Exercise 20.1: A sign graph and its five limiting behaviors

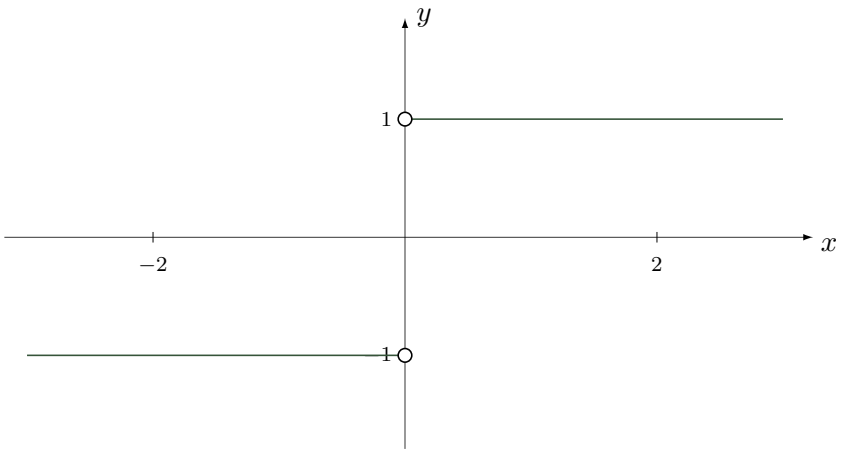
What we must show. Sketch $f(x) = x/|x|$ and inspect the five limits requested in the exercise, in their stated order.

Separate the positive and negative halves of the domain.

The denominator is zero at $x = 0$, so the domain is $\mathbb{R} \setminus \{0\}$. For $x > 0$, $|x| = x$, whereas for $x < 0$, $|x| = -x$. Consequently

$$f(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$$

The graph consists of two horizontal half-lines, with the endpoints at $(0, 1)$ and $(0, -1)$ omitted.



Read the limits, keeping the two sides separate.

approach	$+\infty$	0^+	0^-	$-\infty$	0
limit	1	1	-1	-1	does not exist

The right-hand values are constantly 1 and the left-hand values are constantly -1. Since these two one-sided limits differ, Theorem 20.10 excludes a two-sided limit. Detailed verification of the assertions is given in Exercise 20.5.

Conclusion. The five answers are $1, 1, -1, -1$, and nonexistence.

20.2 Exercise 20.2: Two halves of a quadratic graph

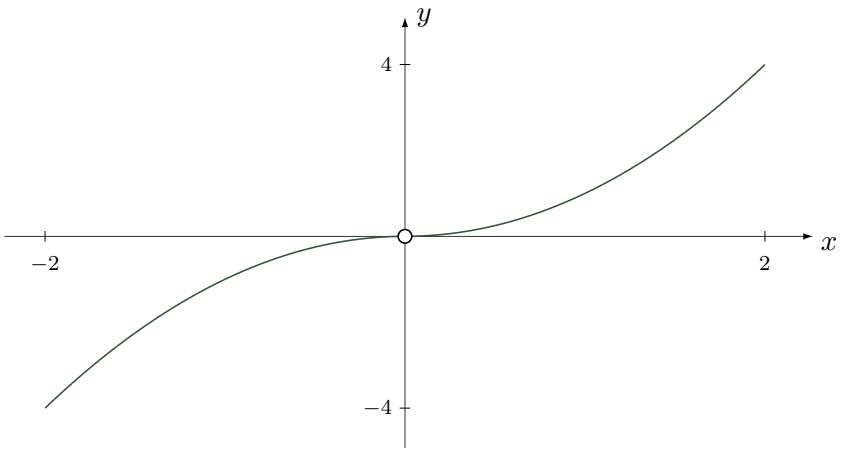
What we must show. Repeat the five-limit inspection for $f(x) = x^3/|x|$.

Cancel only where the denominator is nonzero.

For $x > 0$, $x^3/|x| = x^2$. For $x < 0$, division by $-x$ gives $x^3/|x| = -x^2$. Thus

$$f(x) = \begin{cases} x^2, & x > 0, \\ -x^2, & x < 0. \end{cases}$$

The right branch rises to $+\infty$, the left branch falls to $-\infty$ as x runs toward the corresponding infinity, and both branches approach the missing point $(0, 0)$.



approach	$+\infty$	0^+	0^-	$-\infty$	0
limit	$+\infty$	0	0	$-\infty$	0

The two-sided limit is 0 even though $f(0)$ is undefined.

Conclusion. The five answers are $+\infty, 0, 0, -\infty, 0$. Exercise 20.6 proves them.

20.3 Exercise 20.3: The sine quotient and its removable hole

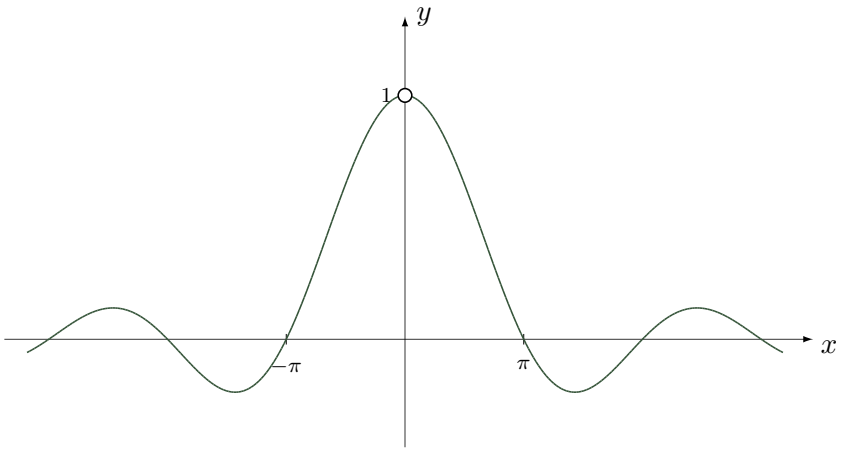
What we must show. Sketch $f(x) = \sin x/x$, $x \neq 0$, and inspect the five limits.

Identify the hole, symmetry, and distant oscillations.

Since $\sin(-x) = -\sin x$, division by $-x$ shows that $f(-x) = f(x)$. The graph is even. Its zeros are the nonzero integer multiples of π, and

$$|f(x)| \leq \frac{1}{|x|} \qquad (x \neq 0).$$

The oscillations therefore shrink at either infinity. Near zero, the curve approaches height 1, with $(0, 1)$ omitted.



approach	$+\infty$	0^+	0^-	$-\infty$	0
limit	0	1	1	0	1

Conclusion. The five limits are $0, 1, 1, 0, 1$. Exercise 20.7 supplies the estimates and the proof near zero.

20.4 Exercise 20.4: Oscillations at zero and a horizontal asymptote

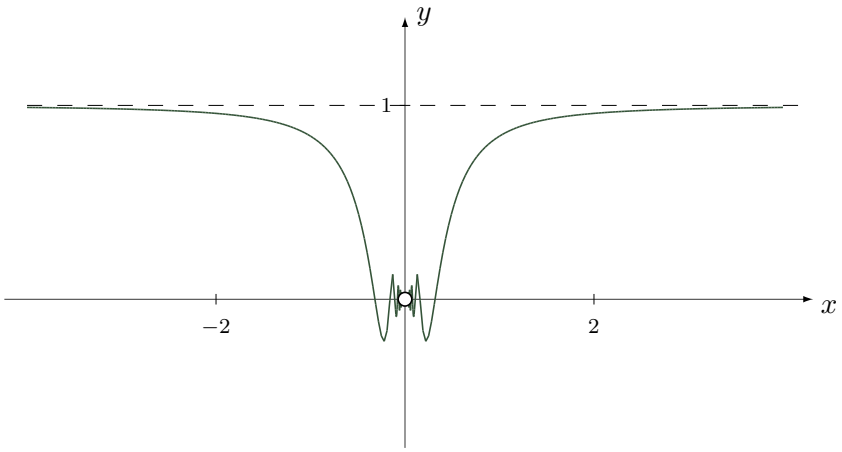
What we must show. Repeat the inspection for $f(x) = x\sin(1/x)$, $x \neq 0$.

Describe both scales of the graph.

The function is even, since $(-x)\sin(-1/x) = x\sin(1/x)$. Its zeros are $x = 1/(k\pi)$ for nonzero integers k. Near zero they accumulate, but

$$|x\sin(1/x)| \leq |x|\,,$$

so the entire oscillating graph is squeezed into a narrowing pair of straight lines. The origin is a hole. At either infinity put $u = 1/x$; then $x\sin(1/x) = \sin u/u$ approaches 1.



The dashed horizontal line is $y = 1$. Only finitely many of the oscillations near the missing origin can be displayed.

approach	$+\infty$	0^+	0^-	$-\infty$	0
limit	1	0	0	1	0

Conclusion. The five answers are $1, 0, 0, 1, 0$. Exercise 20.8 proves both the zero and infinity assertions.

20.5 Exercise 20.5: Proof of the sign-function limit assertions

What we must show. Prove every limit and nonexistence assertion in Exercise 20.1.

The two positive-side limits.

For $x > 0$, $f(x) = 1$, so $|f(x) - 1| = 0 < \varepsilon$ for every $\varepsilon > 0$. For the right-hand limit at zero, any $\delta > 0$ works, because $0 < x < \delta$ implies $x > 0$. For the limit at $+\infty$, the threshold $x > 0$ works for every ε. Thus these limits both equal 1.

The two negative-side limits.

For $x < 0$, $f(x) = -1$, so $|f(x) - (-1)| = 0$. For the left-hand limit at zero, any $\delta > 0$ works whenever $-\delta < x < 0$. For the limit at $-\infty$, all $x < 0$ already have the required value. Thus these two limits equal -1.

Why zero has no two-sided limit of any kind.

The one-sided limits are distinct finite numbers. By Theorem 20.10 a two-sided limit, including an infinite limit, would have to agree with both. It cannot. Equivalently, the sequences $x_n = 1/n$ and $y_n = -1/n$ both approach zero, but their function values are constantly 1 and -1, respectively.

Conclusion. All four one-sided or infinity limits hold, and the two-sided limit at zero does not exist.

20.6 Exercise 20.6: Proof of the signed-quadratic limit assertions

What we must show. Prove all five assertions in Exercise 20.2.

The absolute value controls every approach to zero.

For $x \neq 0$, the piecewise formula gives $|f(x)| = x^2$. Given $\varepsilon > 0$, take $\delta = \sqrt{\varepsilon} > 0$. Then

$$0 < |x| < \delta \quad \Longrightarrow \quad |f(x) - 0| = x^2 < \delta^2 = \varepsilon.$$

This is the two-sided limit 0. The same implication restricted to $x > 0$ or $x < 0$ proves the corresponding one-sided limits.

The two infinities have different signs.

Fix $M > 0$. If $x > \sqrt{M}$, then $x > 0$ and $f(x) = x^2 > M$. Hence $f(x) \to +\infty$ as $x \to +\infty$. If $x < -\sqrt{M}$, then $x < 0$, $x^2 > M$, and $f(x) = -x^2 < -M$. This is $f(x) \to -\infty$ as $x \to -\infty$.

Conclusion. The three limits at zero are 0; the limits at positive and negative infinity are $+\infty$ and $-\infty$, respectively.

20.7 Exercise 20.7: Proof of the sine-quotient limit assertions

What we must show. Justify all five limits in Exercise 20.3.

At either infinity, use a bound that does not depend on phase.

For $x \neq 0$, $|\sin x/x| \leq 1/|x|$. Given $\varepsilon > 0$, if $|x| > 1/\varepsilon$, then

$$\left|\frac{\sin x}{x} - 0\right| \leq \frac{1}{|x|} < \varepsilon.$$

Taking either $x > 1/\varepsilon$ or $x < -1/\varepsilon$ proves the limits 0 at $+\infty$ and $-\infty$.

At zero, justify the familiar sine limit.

We use the Mean Value Theorem, already permitted in Section 19, and the familiar identities $(\sin t)' = \cos t$ and $\sin 0 = 0$. For every nonzero x, the theorem on the interval with endpoints 0 and x gives a point c_x strictly between these endpoints such that

$$\frac{\sin x - \sin 0}{x - 0} = \cos c_x, \qquad |c_x| < |x|\,.$$

The hypotheses hold because sine is continuous on that closed interval and differentiable on its interior.

Let $\varepsilon > 0$. Continuity of cosine at zero, and $\cos 0 = 1$, give $\eta > 0$ such that $|t| < \eta$ implies $|\cos t - 1| < \varepsilon$. If $0 < |x| < \eta$, then $|c_x| < \eta$, so

$$\left|\frac{\sin x}{x} - 1\right| = |\cos c_x - 1| < \varepsilon.$$

This proves the two-sided limit 1. Restricting the same estimate to positive or negative x proves both one-sided limits. It also verifies directly the continuous extension discussed in Section 19, Example 9.

Conclusion. The two infinity limits are 0 and all three limits at zero equal 1.

20.8 Exercise 20.8: Proof for the oscillating product

What we must show. Prove the five assertions in Exercise 20.4.

At zero, the small prefactor controls every oscillation.

The bound $|\sin t| \leq 1$ gives

$$|x \sin(1/x) - 0| \leq |x| \qquad (x \neq 0).$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $0 < |x| < \delta$ forces the displayed error to be less than ε. Hence the two-sided limit is 0, and so are its two one-sided restrictions.

At infinity, transform the expression rather than its factors separately.

For $u = 1/x$, one has $u \neq 0$ and

$$x \sin(1/x) = \frac{\sin u}{u}.$$

Exercise 20.7 proved that for every $\varepsilon > 0$ there is $\eta > 0$ with

$$0 < |u| < \eta \quad \Longrightarrow \quad \left|\frac{\sin u}{u} - 1\right| < \varepsilon.$$

If $|x| > 1/\eta$, then $0 < |1/x| < \eta$, so this bound applies. It works for positive and negative x, proving a limit of 1 at both infinities. Multiplying the separate informal limits $x \to \infty$ and $\sin(1/x) \to 0$ would not be a valid finite product-limit calculation; the quotient identity supplies the needed control.

Conclusion. All three zero limits equal 0 and both infinity limits equal 1.

20.9 Exercise 20.9: A reciprocal term and a slant asymptote

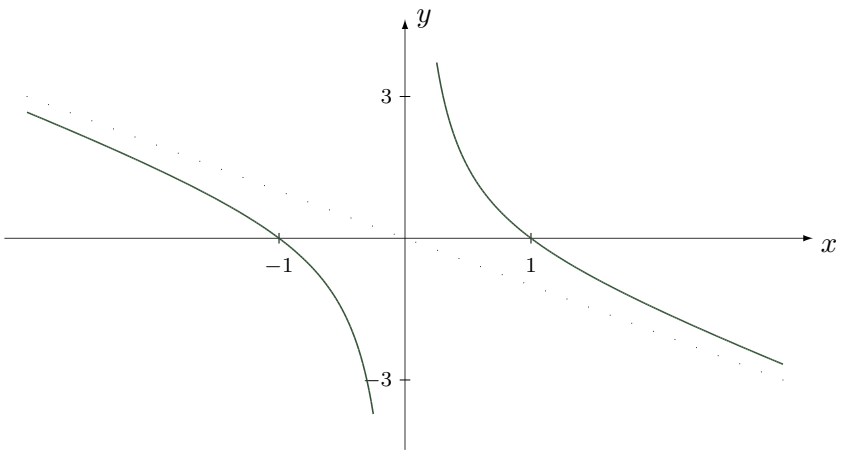
What we must show. Sketch $f(x) = (1 - x^2)/x$ and inspect the same five limits.

Rewrite the function and identify the branches.

The domain excludes zero, and for $x \neq 0$,

$$f(x) = \frac{1}{x} - x.$$

The graph is odd, since $f(-x) = -f(x)$. It crosses the x-axis at $x = \pm 1$. The line $x = 0$ is a vertical asymptote, while $y = -x$ is a slant asymptote because $f(x) - (-x) = 1/x \to 0$ at either infinity.



The dotted diagonal is $y = -x$. The vertical axis is the other asymptote; neither branch contains a point at $x = 0$.

approach	$+\infty$	0^+	0^-	$-\infty$	0
limit	$-\infty$	$+\infty$	$-\infty$	$+\infty$	does not exist

Conclusion. The answers are $-\infty, +\infty, -\infty, +\infty$, and nonexistence. Exercise 20.10 proves them.

20.10 Exercise 20.10: Proof for the reciprocal-minus-linear function

What we must show. Prove every limit assertion in Exercise 20.9.

At positive infinity, the negative linear term dominates.

Fix $M > 0$, and choose $R = \max\{1, M + 1\}$. If $x > R$, then $1/x < 1$ and $x > M + 1$. Consequently

$$f(x) = \frac{1}{x} - x < 1 - x < -M.$$

This proves the limit $-\infty$ at positive infinity. For $x < -R$, put $t = -x > R$. Oddness gives $f(x) = f(-t) = -f(t) > M$. Hence the limit at negative infinity is $+\infty$.

At zero, the reciprocal term dominates.

For the same arbitrary $M > 0$, choose $\delta = \min\{1, 1/(M+1)\} > 0$. If $0 < x < \delta$, then $1/x > M + 1$ and $x < 1$, giving

$$f(x) = \frac{1}{x} - x > (M+1) - 1 = M.$$

This proves the right-hand limit $+\infty$ at zero. If $-\delta < x < 0$, put $t = -x \in (0, \delta)$. Then $f(x) = -f(t) < -M$, proving the left-hand limit $-\infty$. The one-sided limits differ, so Theorem 20.10 shows that the two-sided limit does not exist, even as an infinite limit.

Conclusion. All five assertions follow from explicit threshold estimates, with the signs preserved on each side of zero.

20.11 Exercise 20.11: Three removable algebraic quotients

What we must show. Find the three limits by simplifying only at points where the original quotients are defined.

(a) Factor the difference of squares.

For $x \neq a$,

$$\frac{x^2 - a^2}{x - a} = \frac{(x-a)(x+a)}{x-a} = x + a.$$

The expression on the right is continuous at a. Since the punctured expressions agree, their limits agree, and the answer is $a + a = 2a$. No value of the original quotient at a is needed.

(b) Multiply by the conjugate, with the domain checked.

Here $b > 0$. For example, $|x - b| < b/2$ ensures $x > b/2 > 0$, so both square roots exist. If also $x \neq b$, then

$$\frac{\sqrt{x} - \sqrt{b}}{x - b} = \frac{(\sqrt{x} - \sqrt{b})(\sqrt{x} + \sqrt{b})}{(x-b)(\sqrt{x} + \sqrt{b})} = \frac{1}{\sqrt{x} + \sqrt{b}}.$$

The conjugate is positive, and the cancellation uses $x - b \neq 0$. As $x \to b$, continuity of the square-root function gives a denominator tending to $2\sqrt{b} > 0$. The reciprocal limit is therefore $1/(2\sqrt{b})$.

(c) Factor the difference of cubes.

The identity $x^3 - a^3 = (x-a)(x^2 + ax + a^2)$ follows by multiplying out; the two mixed terms cancel. For $x \neq a$ it gives

$$\frac{x^3 - a^3}{x - a} = x^2 + ax + a^2.$$

This polynomial is continuous at a, so the limit is $a^2 + a^2 + a^2 = 3a^2$.

Conclusion. The limits in (a), (b), and (c) are $2a$, $1/(2\sqrt{b})$, and $3a^2$, respectively.

20.12 Exercise 20.12: One simple pole and one squared pole

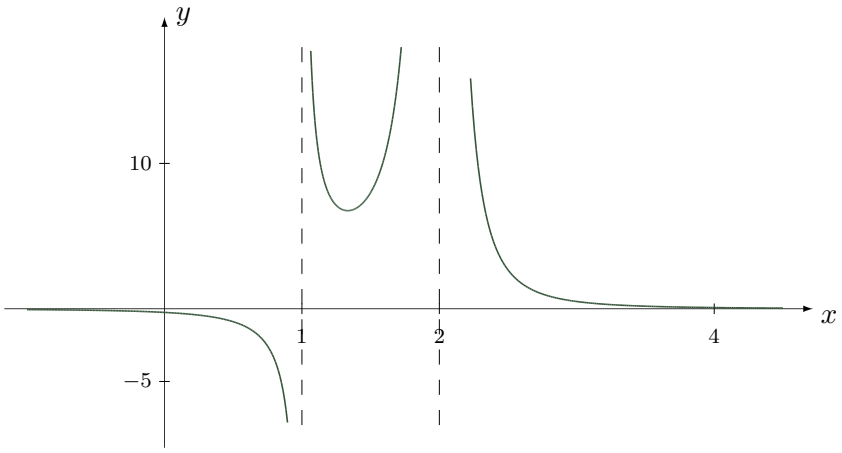
What we must show. Sketch $f(x) = 1/[(x-1)(x-2)^2]$, determine all four requested one-sided limits, and decide the two two-sided limits.

(a) Use signs and poles to organize the sketch.

The domain is $\mathbb{R} \setminus \{1, 2\}$. For $x \neq 2$, $(x-2)^2 > 0$, so the sign of $f(x)$ is the sign of $x - 1$. The graph is negative to the left of 1 and positive on both $(1, 2)$ and $(2, \infty)$. There are vertical asymptotes at 1 and 2 and a horizontal asymptote $y = 0$ at both infinities. For additional shape information, differentiation on each domain interval gives

$$f'(x) = -\frac{3x-4}{(x-1)^2(x-2)^3}.$$

Thus the middle positive branch decreases on $(1, 4/3)$ and increases on $(4/3, 2)$, reaching $f(4/3) = 27/4$. The left branch decreases, and the branch to the right of 2 decreases to zero. These derivative facts concern only the sketch; the limit estimates below do not require them.



The dashed vertical lines are $x = 1$ and $x = 2$. The portions outside the finite plotting window continue toward the indicated infinite limits.

(b), near 2: the squared factor is positive on both sides.

If $0 < |x-2| < 1/2$, then $1/2 < x - 1 < 3/2$. Hence

$$f(x) = \frac{1}{(x-1)(x-2)^2} > \frac{2}{3\,|x-2|^2} > 0.$$

Given $M > 0$, choose $\delta = \min\{1/2, \sqrt{2/(3M)}\}$. Then $0 < |x-2| < \delta$ makes this last lower bound exceed M. Thus both $\lim_{x\to 2^+} f(x)$ and $\lim_{x\to 2^-} f(x)$ equal $+\infty$.

(b), near 1: the unsquared factor changes sign.

If $0 < |x-1| < 1/2$, then $1/2 < |x-2| < 3/2$, so

$$|f(x)| = \frac{1}{|x-1|\,(x-2)^2} > \frac{4}{9\,|x-1|}.$$

For $M > 0$, taking $\delta = \min\{1/2, 4/(9M)\}$ makes the magnitude exceed M. For $x > 1$ the function is positive, so the right-hand limit is $+\infty$. For $x < 1$ it is negative, so the left-hand limit is $-\infty$.

(c) Compare the two sides at each pole.

At 2 the two sides agree, giving $\lim_{x\to 2} f(x) = +\infty$ by Theorem 20.10. At 1 they disagree, so $\lim_{x\to 1} f(x)$ does not exist.

Conclusion. In the requested order the one-sided limits are $+\infty, +\infty, +\infty, -\infty$. The two-sided limit at 2 is $+\infty$; the one at 1 does not exist.

20.13 Exercise 20.13: Finite limit laws with reciprocal and square root

What we must show. From $\lim_{x\to a} f(x) = 3$ and $\lim_{x\to a} g(x) = 2$, prove all three asserted limits.

(a) Apply the finite sum and product laws.

The limits are finite, so Theorem 20.4 gives

$$\lim_{x\to a} g(x)^2 = 2 \cdot 2 = 4, \qquad \lim_{x\to a} 3f(x) = 3 \cdot 3 = 9.$$

Adding these limits gives

$$\lim_{x\to a}[3f(x) + g(x)^2] = 9 + 4 = 13.$$

(b) Ensure the reciprocal is defined before using its law.

Since $g(x) \to 2$, there is a punctured neighborhood of a on which $|g(x) - 2| < 1$. There $g(x) > 1$, so $g(x) \neq 0$. On that neighborhood the quotient law applies, with the constant numerator 1 and denominator limit $2 \neq 0$. Therefore

$$\lim_{x\to a} \frac{1}{g(x)} = \frac{1}{2}.$$

(c) Check the square-root domain near the limiting point.

The sum and scalar laws first give

$$h(x) = 3f(x) + 8g(x) \longrightarrow 3 \cdot 3 + 8 \cdot 2 = 25.$$

For all sufficiently close $x \neq a$, one has $|h(x) - 25| < 1$, hence $h(x) > 24 > 0$. Thus the square root is defined throughout a punctured neighborhood. The square-root function is continuous at 25; Theorem 20.5, applied on this neighborhood, gives

$$\lim_{x\to a} \sqrt{3f(x) + 8g(x)} = \sqrt{25} = 5.$$

Conclusion. The three limits are 13, 1/2, and 5. Both nonzero-denominator and nonnegative-radicand conditions hold on a sufficiently small punctured neighborhood.

20.14 Exercise 20.14: The two one-sided infinite reciprocal limits

What we must show. Prove $\lim_{x\to 0^+} 1/x = +\infty$ and $\lim_{x\to 0^-} 1/x = -\infty$ directly.

Right side: choose the distance from the target height.

Fix $M > 0$ and put $\delta = 1/M > 0$. If $0 < x < \delta$, reciprocation of positive numbers reverses the strict inequality and yields $1/x > 1/\delta = M$. This is precisely the threshold definition of the first limit.

Left side: keep track of the negative sign.

For the same M and δ, if $-\delta < x < 0$, then $0 < -x < \delta$. The previous positive estimate gives $1/(-x) > M$. Multiplying by -1 reverses the inequality: $1/x < -M$. Since every negative threshold is $-M$ for some $M > 0$, this proves the second limit.

Conclusion. The reciprocal tends to $+\infty$ from the right and to $-\infty$ from the left.

20.15 Exercise 20.15: Two limits of the shifted reciprocal cube

What we must show. For the function in Example 4, $f(x) = 1/(x-2)^3$ with $x \neq 2$, prove the limits at $-\infty$ and 2^+ requested in the exercise.

At negative infinity, bound the absolute error from zero.

Let $\varepsilon > 0$ and set $A = -\varepsilon^{-1/3} < 0$. If $x < A$, then $x < 0$ and $|x-2| = 2 - x > -x > \varepsilon^{-1/3}$. Cubing positive quantities and then reciprocating gives

$$|f(x) - 0| = \frac{1}{|x-2|^3} < \varepsilon.$$

This proves $\lim_{x\to-\infty} f(x) = 0$.

At the pole from the right, choose a small positive denominator.

Given $M > 0$, put $\delta = M^{-1/3} > 0$. If $2 < x < 2 + \delta$, then $0 < x - 2 < \delta$. Hence $(x-2)^3 < \delta^3 = 1/M$ and $f(x) = 1/(x-2)^3 > M$. Thus $\lim_{x\to 2^+} f(x) = +\infty$.

Conclusion. The two limits are 0 and $+\infty$, respectively.

20.16 Exercise 20.16: Order of right-hand limits and the loss of strictness

What we must show. Prove the order assertion in part (a), and determine whether strict pointwise order implies strict order of limits in part (b). The existing limits may be finite or infinite.

(a) Approach through one common sequence.

Shrink $b > a$ if necessary so that the interval (a, b) lies in the neighborhoods used to define both right-hand limits. Let

$$x_n = a + \frac{b-a}{n+1}.$$

Then $a < x_n < b$ and $x_n \to a$. The hypotheses therefore give

$$f_1(x_n) \longrightarrow L_1, \qquad f_2(x_n) \longrightarrow L_2, \qquad f_1(x_n) \le f_2(x_n) \quad (n \in \mathbb{N}).$$

The order rule for existing sequence limits, Exercise 9.9(c), now gives $L_1 \le L_2$, including the possible infinite values. To recall why that order rule includes these values, if $L_1 > L_2$, choose a finite real c strictly between them. Convergence would eventually give both $f_1(x_n) > c$ and $f_2(x_n) < c$, contrary to the pointwise inequality.

(b) Give a counterexample with strictly ordered values.

On $(a, a+1)$ put $f_1(x) = 0$ and $f_2(x) = x - a$. Every point satisfies $f_1(x) < f_2(x)$, since $x > a$. Nevertheless both right-hand limits at a equal 0. The positive gap $x - a$ shrinks to zero, so no strict gap remains between the limits.

Conclusion. Part (a) gives $L_1 \le L_2$. The answer to part (b) is no: equality of limits is possible despite strict inequality at every point.

20.17 Exercise 20.17: The squeeze lemma proves existence of the middle limit

What we must show. Suppose $f_1(x) \le f_2(x) \le f_3(x)$ on (a, b), and both outside functions have right-hand limit L at a. Prove that f_2 has that same limit, without assuming it exists.

Finite L: enclose all middle values in a prescribed error interval.

Let $\varepsilon > 0$. The two given limits supply $\delta_1, \delta_3 > 0$ such that

$$\begin{aligned} a < x < a + \delta_1 \quad &\Longrightarrow \quad L - \varepsilon < f_1(x) < L + \varepsilon, \\ a < x < a + \delta_3 \quad &\Longrightarrow \quad L - \varepsilon < f_3(x) < L + \varepsilon. \end{aligned}$$

Take $\delta = \min\{\delta_1, \delta_3, b - a\} > 0$. Whenever $a < x < a + \delta$, the point lies in (a, b) and

$$L - \varepsilon < f_1(x) \le f_2(x) \le f_3(x) < L + \varepsilon.$$

Thus $|f_2(x) - L| < \varepsilon$. This verifies the definition of its limit and establishes existence as part of the conclusion.

Infinite L: one of the outside functions already forces the threshold.

If $L = +\infty$, let $M > 0$. The limit of f_1 gives $\delta_1 > 0$ such that $a < x < a + \delta_1$ implies $f_1(x) > M$. Restrict also to $x < b$. Then $f_2(x) \ge f_1(x) > M$, proving $f_2(x) \to +\infty$. If $L = -\infty$, let $M > 0$ and use the upper function instead: eventually $f_3(x) < -M$, so $f_2(x) \le f_3(x) < -M$. This proves $f_2(x) \to -\infty$.

Conclusion. The middle limit exists and equals L in every finite or infinite case. No order theorem requiring its prior existence was used.

20.18 Exercise 20.18: Rationalizing a square-root difference

What we must show. Show the limit at zero exists for

$$f(x) = \frac{\sqrt{1 + 3x^2} - 1}{x^2} \qquad (x \neq 0),$$

and determine its value.

Check every factor used in the rationalization.

For every real x, $1+3x^2 \geq 1$, so the square root exists and $\sqrt{1+3x^2}+1 \geq 2 > 0$. If $x \neq 0$, then $x^2 > 0$. We may therefore multiply the numerator and denominator by the conjugate and cancel the nonzero factor x^2:

$$\begin{aligned} f(x) &= \frac{(\sqrt{1+3x^2}-1)(\sqrt{1+3x^2}+1)}{x^2(\sqrt{1+3x^2}+1)} \\ &= \frac{(1+3x^2)-1}{x^2(\sqrt{1+3x^2}+1)} = \frac{3}{\sqrt{1+3x^2}+1}. \end{aligned}$$

Use a continuous expression that also makes sense at zero.

The polynomial $1+3x^2$ tends to 1 as $x \to 0$. Continuity of square root gives a denominator tending to $\sqrt{1}+1 = 2 \neq 0$. The reciprocal and scalar limit laws yield

$$\lim_{x\to 0} f(x) = \frac{3}{2}.$$

This establishes existence, not just a possible value: the original function agrees on every nonzero point with an expression continuous at zero.

Conclusion. The limit exists and equals 3/2.

20.19 Exercise 20.19: Changing the right-hand approach interval

What we must show. Let $a < b_1 < b_2$ and let f be defined on (a, b_2). Show that existence and value of the limit along (a, b_i) are independent of choosing $i = 1$ or $i = 2$.

A limit along the larger interval restricts to the smaller one.

Suppose the limit along $S_2 = (a, b_2)$ exists and equals L. Every sequence in $S_1 = (a, b_1)$ is also a sequence in S_2, since $S_1 \subseteq S_2$. Thus whenever a sequence in S_1 tends to a, its function values tend to L by the given limit. This is the sequential definition of the limit along S_1, with the same value.

A sequence in the larger interval eventually enters the smaller one.

Conversely, suppose the limit along S_1 exists and equals L. Take an arbitrary sequence (x_n) in S_2 tending to a. The positive number $b_1 - a$ is an allowable tolerance for convergence. Therefore for some integer N,

$$n > N \quad \Longrightarrow \quad |x_n - a| < b_1 - a.$$

Since all these $x_n > a$, the inequality says $a < x_n < b_1$. Hence the tail $x_{N+1}, x_{N+2}, \ldots$ is a sequence in S_1 and still tends to a. By the assumed smaller-interval limit, the corresponding function values tend to L. Adding back finitely many initial terms does not change a finite or infinite sequence limit. Thus $f(x_n) \to L$ for the whole arbitrary sequence, proving the limit along S_2.

Both approach sets admit sequences tending to a, for example $a + (b_1 - a)/(n+1)$, so their limits are meaningful under Definition 20.1 and are unique. The proof did not require L to be finite.

Conclusion. Either interval defines the same right-hand limit, finite or infinite, which is denoted $\lim_{x\to a^+} f(x)$.

20.20 Exercise 20.20: Infinite sum and product limit rules

What we must show. Assume $\lim_{x\to a^S} f_1(x) = +\infty$ and that $L_2 = \lim_{x\to a^S} f_2(x)$ exists. Prove all three asserted rules and analyze the zero case in part (d). Here a may be finite or infinite, with the approach set S as in Definition 20.1.

Use an arbitrary admissible sequence, so all approach types are covered.

Take any sequence $x_n \in S$ with $x_n \to a$, and write $u_n = f_1(x_n)$, $v_n = f_2(x_n)$. The hypotheses give $u_n \to +\infty$ and $v_n \to L_2$. We prove each conclusion for this arbitrary sequence. Definition 20.1 will then give the corresponding function limit.

(a) A finite or positive-infinite second limit cannot cancel the first.

Suppose $L_2 \neq -\infty$. There is a finite real lower bound c for all sufficiently late v_n: if L_2 is finite, take $c = L_2 - 1$; if $L_2 = +\infty$, take $c = 0$. For any $M > 0$, the limit $u_n \to +\infty$ eventually gives $u_n > M - c$ (or a larger positive threshold if needed). On the common tail,

$$u_n + v_n \geq u_n + c > M.$$

Hence $u_n + v_n \to +\infty$, so $\lim_{x\to a^S}(f_1 + f_2)(x) = +\infty$.

(b) A positive second limit gives a fixed positive factor.

Suppose $0 < L_2 \leq +\infty$. If L_2 is finite, take $c = L_2/2 > 0$. If $L_2 = +\infty$, take $c = 1$. In either case $v_n \geq c$ eventually. For any $M > 0$, eventually $u_n > M/c > 0$ as well. Multiplication by this positive u_n preserves the inequality, giving

$$u_n v_n \geq c u_n > M.$$

Thus $u_n v_n \to +\infty$, proving the asserted product rule.

(c) Reflect a negative second limit before applying the positive rule.

Suppose $-\infty \leq L_2 < 0$. Then $-v_n \to -L_2 > 0$, where $-L_2 = +\infty$ if $L_2 = -\infty$. Part (b)'s argument applied to u_n and $-v_n$ gives $u_n(-v_n) \to +\infty$. Negating reverses every threshold inequality, so $u_n v_n \to -\infty$. Since the admissible sequence was arbitrary, this proves $\lim_{x\to a^S}(f_1 f_2)(x) = -\infty$.

(d) A zero second limit does not determine the product.

Use the common approach $x \to 0^+$ on $S = (0, 1)$, and fix $f_1(x) = 1/x^2$, which tends to $+\infty$. For any prescribed real c, choose $f_2(x) = cx^2$. Then $f_2(x) \to 0$ but $f_1(x)f_2(x) = c$, so any finite real limit can occur. If $f_2(x) = x$, its limit is again 0 but the product is $1/x \to +\infty$. If $f_2(x) = -x$, the product tends to $-\infty$.

There need not be any product limit: choose $f_2(x) = x^2 \sin(1/x)$. The bound $|f_2(x)| \leq x^2$ gives $f_2(x) \to 0$, while the product is $\sin(1/x)$. Along

$$x_n = \frac{1}{2\pi n + \pi/2}, \qquad y_n = \frac{1}{2\pi n + 3\pi/2},$$

both of which belong to $(0, 1)$ and tend to zero, its values are 1 and -1, respectively. These incompatible sequence limits exclude a function limit.

Conclusion. Parts (a), (b), and (c) give $+\infty$, $+\infty$, and $-\infty$, respectively. When $L_2 = 0$, the product may have any finite limit, either infinite limit, or no limit.